

Name: _____

GCSE Chemistry Unit 2 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries

Year	E	D	C	B	A	A*	Max
2018	7	10	16	27	39	53	67
2019	4	8	16	25	35	46	57
2022	6	8	17	27	38	51	64
2023	6	9	11	20	29	39	49

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE – NEW**

3410UB0-1



S18-3410UB0-1

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

HIGHER TIER

THURSDAY, 17 MAY 2018 – MORNING

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper
 you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen
 or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional
 page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5(a)** is a quality of extended response (QER) question where your writing skills will be
 assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common
 ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	12	
3.	8	
4.	9	
5.	11	
6.	7	
7.	9	
8.	9	
9.	7	
Total	80	

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Answer all questions.

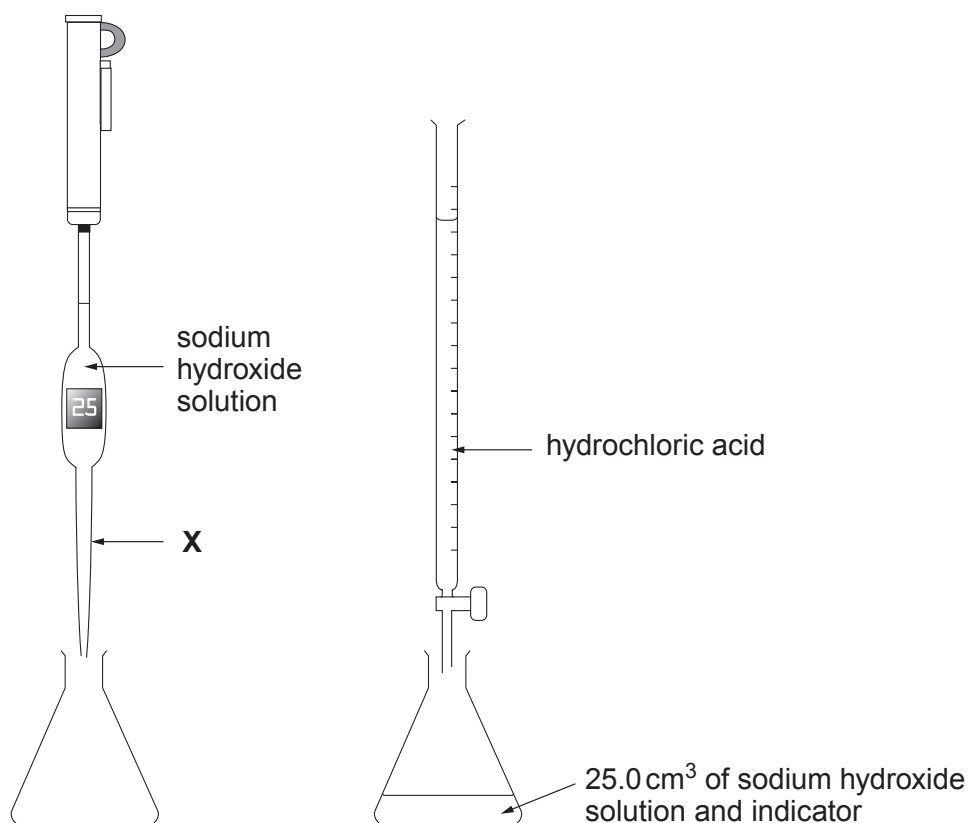
1. (a) Hydrochloric acid, HCl, reacts with sodium hydroxide solution to form sodium chloride and water only.

Write a balanced **symbol** equation for this reaction.

[2]

- (b) A group of students was asked to find the volume of hydrochloric acid solution needed to neutralise 25.0 cm³ of sodium hydroxide solution. They decided to titrate sodium hydroxide with hydrochloric acid.

Apparatus



Results

Titre	1	2	3
Volume of hydrochloric acid needed (cm ³)	18.2	17.8	18.0



- (i) Name the piece of apparatus **X**. [1]

.....

- (ii) Explain the purpose of the indicator. [1]

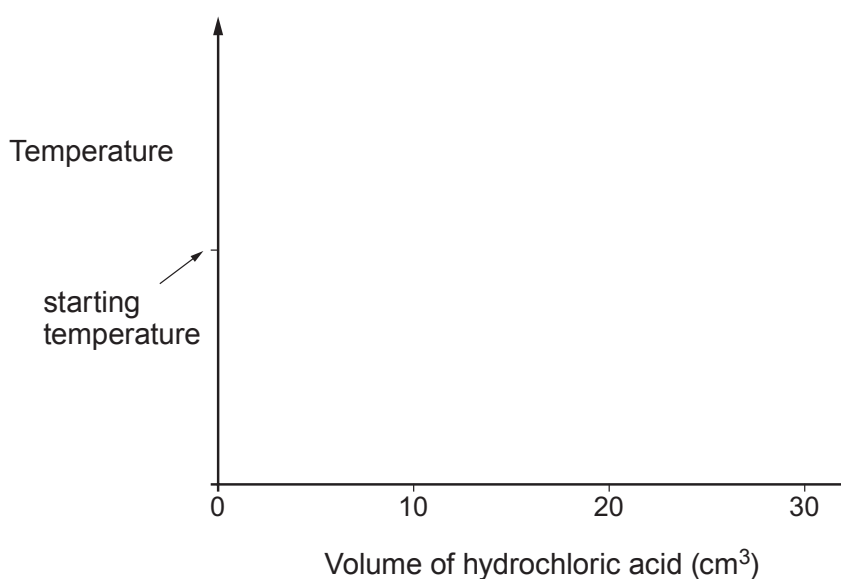
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- (iii) Calculate the mean volume of hydrochloric acid needed to neutralise 25.0 cm^3 of the sodium hydroxide solution. [1]

Mean volume = cm^3

- (iv) The change in temperature during the reaction can be monitored using a temperature sensor.

Sketch a graph on the axes below to show how the temperature changes as more and more acid up to a total of 30 cm^3 is added. [2]



- (v) The experiment was repeated using hydrochloric acid of **half** the original concentration.

State the volume of hydrochloric acid that would be needed to change the indicator colour. [1]

Volume = cm^3



2. (a) When a mixture of iron(III) oxide and aluminium powder (Thermit mixture) is heated, there is a violent reaction. The reaction is carried out in a tube surrounded by a mound of sand because the temperature reaches 2500 °C. A bead of iron is recovered from the sand. The picture below shows the reaction taking place in a darkened room.



- (i) Give the reason why the iron formed in the reaction is molten. [1]

.....
.....

- (ii) Complete and balance the **symbol** equation for this reaction. [2]



- (iii) State which of the substances is oxidised. Give the reason for your choice. [1]

.....
.....

- (iv) When a mixture of magnesium oxide and aluminium powder is heated, there is no reaction.

List iron, magnesium and aluminium in order of reactivity. [1]

Most reactive

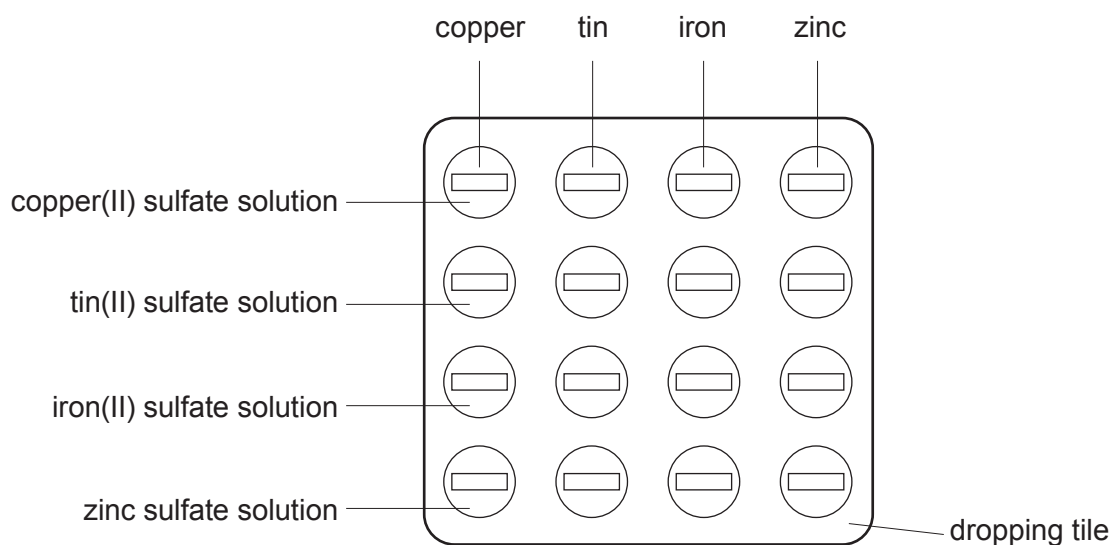
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Least reactive



- (b) Some metals are more reactive than others. A more reactive metal displaces a less reactive metal from its compounds.

A student was given tin, iron, copper and zinc and solutions of the metal sulfates. Using a dropping pipette, she put a little of one of the sulfate solutions in four of the depressions of the dropping tile. She did this for each solution in turn. She then put a piece of metal foil in each of the solutions, as shown below.



- (i) Put a tick (✓) next to the question which **best** describes the investigation the student is carrying out. [1]

Which displacement is the most exothermic?

☐

Which metal can displace copper from solution?

☐

What is meant by the reactivity series?

☐

What are the positions of the four metals in the reactivity series?

☐


- (ii) The student recorded the results by putting a tick (✓) next to a mixture which showed signs of a reaction and a cross (X) next to a mixture which showed no signs of a reaction.

The student concluded that:

tin displaces copper
iron displaces tin
iron displaces copper
zinc displaces iron

Give the **letter** of the tile below which shows the results she recorded.

[1]

Letter

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

A

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

B

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

C

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

D

- (iii) Another student said that not all of the tests were necessary. Give **one** example of a test not needed. Explain your choice.

[2]

Example

Explanation



- (c) Copper displaces silver from a solution of silver nitrate, AgNO_3 , to form copper(II) nitrate solution.

(i) Describe **one** change the student would **see** during this displacement reaction. [1]

.....

.....

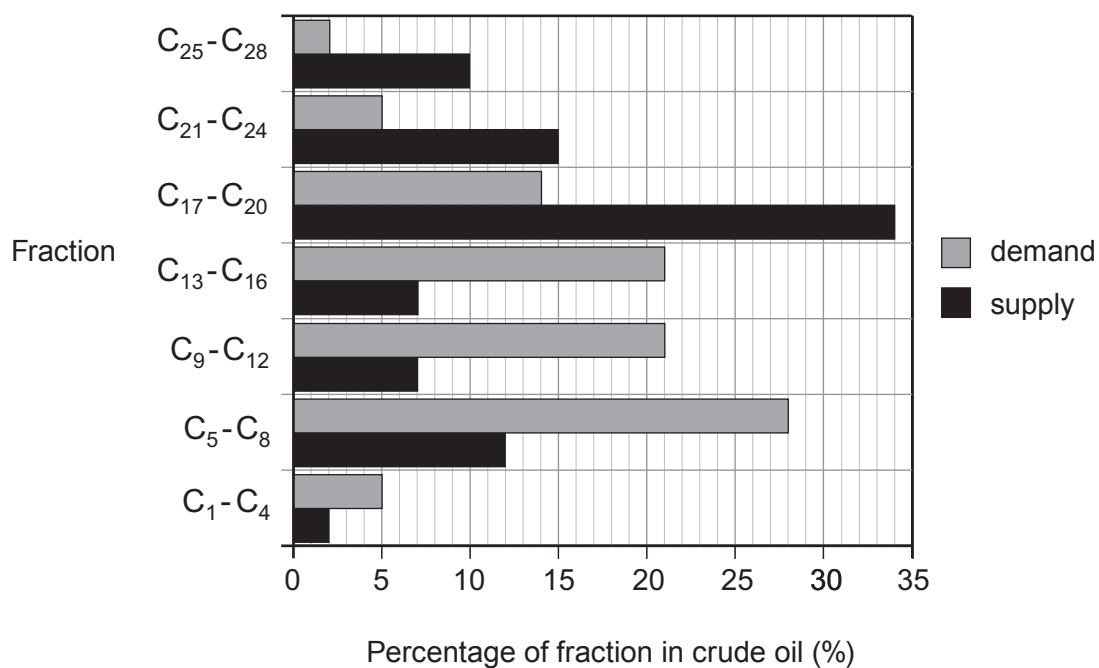
(ii) Write a balanced **symbol** equation for this reaction. [2]

.....

12

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3. (a) The bar chart shows the relative *supply* and *demand* for some fractions obtained from crude oil.



- (i) Use the bar chart to describe how the **difference** between *supply* and *demand* of the fractions changes as chain length increases. [2]

.....

.....

.....

.....

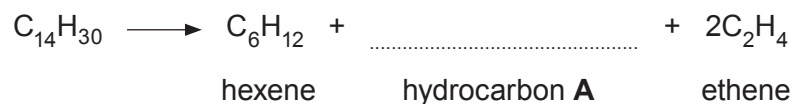
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- (ii) Oil companies have solved the problem of the over-supply of some fractions by using a process called cracking.

$C_{14}H_{30}$ can be cracked forming hexene, ethene and hydrocarbon **A**.

- I. Complete the equation for the cracking of $C_{14}H_{30}$. [1]



- II. Name hydrocarbon **A**. [1]

- III. State why ethene is considered an important raw material. [1]

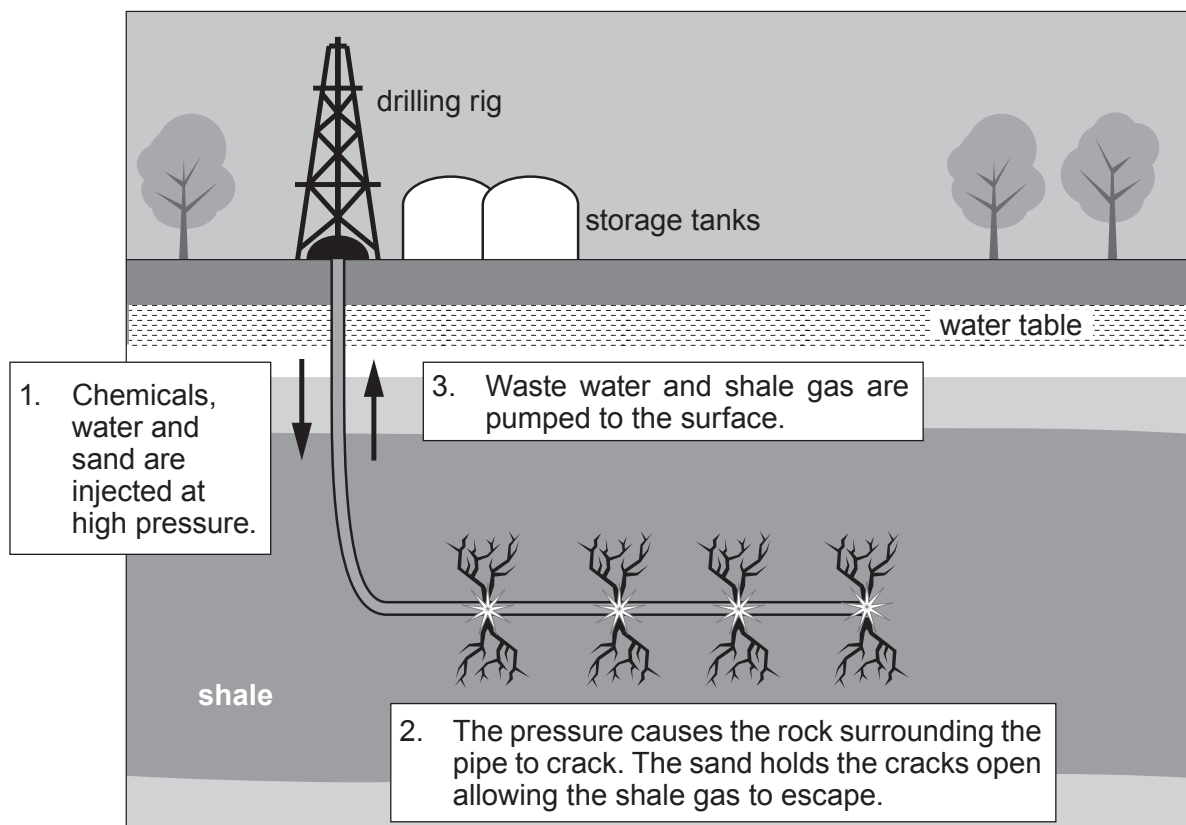
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- (iii) One hydrocarbon found in the C_1 - C_4 fraction is propane. Propane burns in air forming carbon dioxide and water.

Balance the symbol equation that represents this reaction. [1]



- (b) Shale gas is natural gas trapped in rocks deep underground. Like oil and coal, shale gas has, essentially, formed from the remains of plants, animals and micro-organisms that lived millions of years ago. It can be extracted using a process known as hydraulic fracturing – or “fracking” – which involves drilling long horizontal wells into the rocks more than a kilometre below the surface. Massive quantities of water, sand and chemicals are pumped into the wells at high pressure. This opens up cracks in the shale, which are held open by the sand, enabling the trapped gas to escape to the surface for collection.



Supporters of fracking argue that extracting shale gas deposits will help keep energy affordable and cut consumption of dirtier coal. But opponents claim fracking is dangerous and polluting, and that tapping into extra shale gas supplies will increase carbon dioxide emissions. The main controversy surrounding shale gas is the potential of fracking to contaminate drinking water supplies with shale gas or drilling chemicals. Other issues include the huge quantities of water and chemicals used in the extraction process, the waste water generated and possible earthquake tremors.



- (i) Put a tick (✓) in the box next to the statement that identifies the substance(s) recovered during the fracking process. [1]

only shale gas

☐

shale gas and contaminated water

☐

shale gas and sand

☐

shale gas, sand and toxic chemicals

☐

- (ii) Put a tick (✓) in the box next to the statement which is true. [1]

burning shale gas does not cause global warming

☐

chemicals used in fracking are contaminating our drinking water

☐

fracking produces vast quantities of contaminated water

☐

shale gas is a renewable energy source

☐

shale gas is cheaper than other fossil fuels

☐


4. (a) Calcium and oxygen react forming calcium oxide.

(i) Explain, using dot and cross diagrams, how bonding takes place during the formation of calcium oxide. [2]

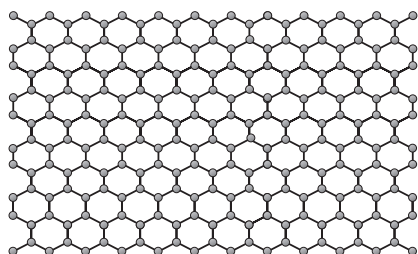
(ii)

Substance	Melting point (°C)
calcium oxide	2613
sodium chloride	801

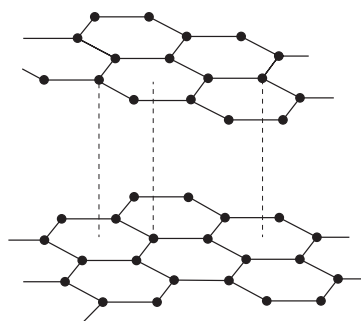
Explain why both of these compounds have high melting points and why the melting point of calcium oxide is higher. [2]



(b) Graphene and graphite are different forms of carbon.



graphene



graphite

(i) Explain how the bonding present in both forms makes them good conductors of electricity. [2]

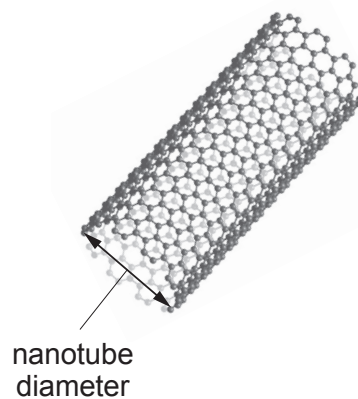
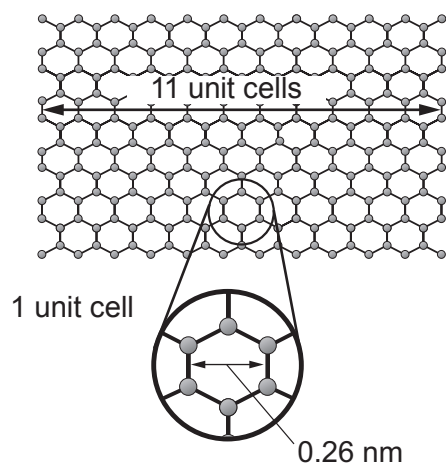
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(ii) Carbon nanotubes can be formed by rolling up graphene sheets.



A sheet of graphene 11 unit cells in width was rolled up to form a nanotube. Use the information below to calculate the diameter, in metres, of the nanotube formed. Write your answer in **standard form**. [3]

$$\text{circumference} = \pi \times \text{diameter}$$

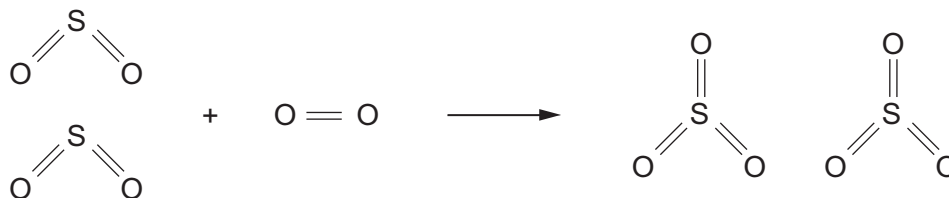
$$\pi = 3.14$$

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

Diameter = m



- (b) The equation shows the bonds which are broken and the bonds which are formed in the production of sulfur trioxide.



The bond energy of an $\text{S}=\text{O}$ bond is 523 kJ.

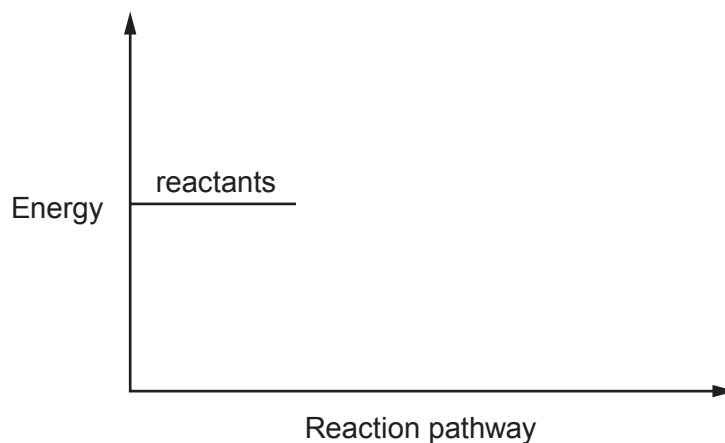
- (i) The **total** energy needed to break the bonds in the reactants is 2587 kJ. Calculate the energy needed to break an $\text{O}=\text{O}$ bond. [2]

Energy = kJ

- (ii) Calculate the overall energy change for the reaction. [2]

Overall energy change = kJ

- (iii) Complete the energy profile for the reaction. [1]



6. (a) Fertilisers usually contain compounds of three essential elements required for healthy and productive plant growth. The table shows the properties of two nitrogenous fertilisers.

Property	Ammonium nitrate	Urea
% nitrogen	34	46
Mass of fertiliser absorbed by plants per 2000 litre spreader (kg)	690	560
Solubility	very soluble	very soluble
Cost (£/tonne)	360	350
Loss of ammonia due to evaporation	low	high
Weather dependency	effective in all conditions	effectiveness dependent on specific temperature and rainfall conditions

- (i) Which fertiliser leads to the absorption of more **nitrogen** from a full 2000 litre spreader? Show your working. [2]

Fertiliser

- (ii) Urea is the most widely used fertiliser in the world. However, many British farmers prefer to use ammonium nitrate.

Put a tick (✓) in the box next to the statement which suggests the **main** reason for this. [1]

ammonium nitrate contains less nitrogen than urea

☐

ammonium nitrate is better suited to British weather conditions than urea

☐

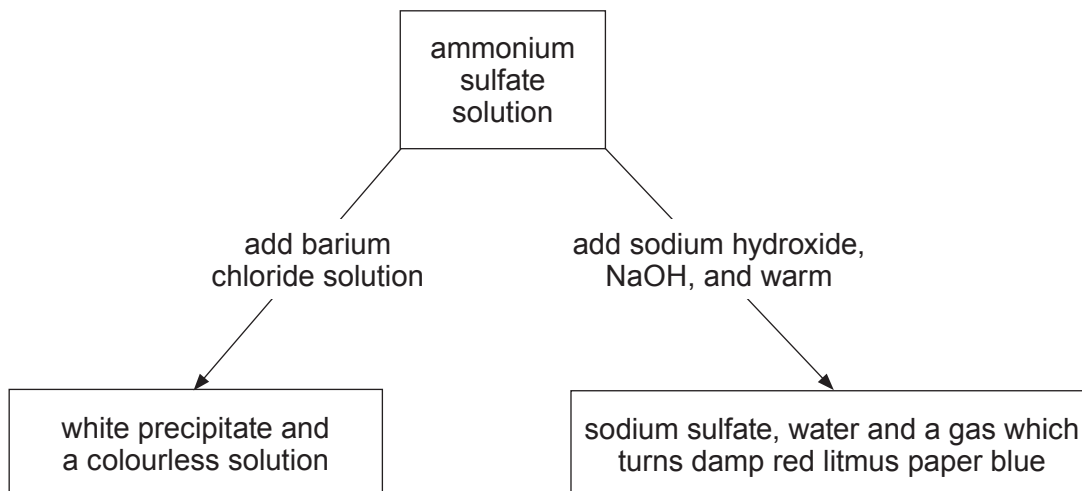
more ammonium nitrate is absorbed by plants than urea

☐

ammonium nitrate is more expensive than urea

☐


- (b) The flow chart shows the chemical tests for the identification of the ions present in ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.



- (i) Write a balanced **symbol** equation for the reaction between ammonium sulfate and sodium hydroxide. [2]

- (ii) The symbol equation below represents the reaction occurring between ammonium sulfate solution and barium chloride solution.



Write the **ionic** equation for the reaction. Include the state symbols. [2]

..... + $\longrightarrow$



7. (a) The table shows the first four members of the alkane family.

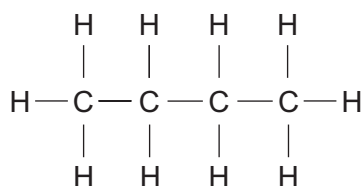
Alkanes
methane, CH ₄
ethane, C ₂ H ₆
propane, C ₃ H ₈
butane, C ₄ H ₁₀

Give the **general** formula for the alkane family.

[1]

.....

- (b) C₄H₁₀ has two isomers. The diagram below shows the structure of one of the isomers, butane.



Draw and name the other isomer.

[2]

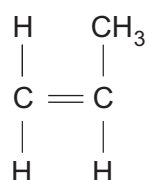
Isomer structure

Name of isomer



(c) The flow diagram shows two reactions of propene.

Draw the structural formula for compound **A** and the repeating unit in polymer **B**. [2]



bromine, Br₂

A

polymerisation

B

(d) A student was given a colourless liquid. He suspects the liquid is ethanol.

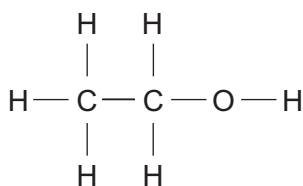
Describe a chemical test that he could carry out to positively identify the liquid as ethanol. Include the observation he would make. [2]

Test

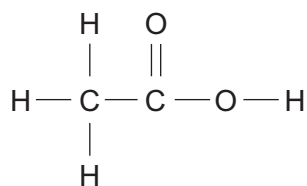
Observation



(e) Organic compounds can also be identified using infrared spectroscopy.



ethanol



ethanoic acid

Bond	Wavenumber (cm^{-1})
$\text{C}=\text{O}$	1650 to 1750
$\text{C}-\text{H}$	2800 to 3100
$\text{O}-\text{H}$	2500 to 3550

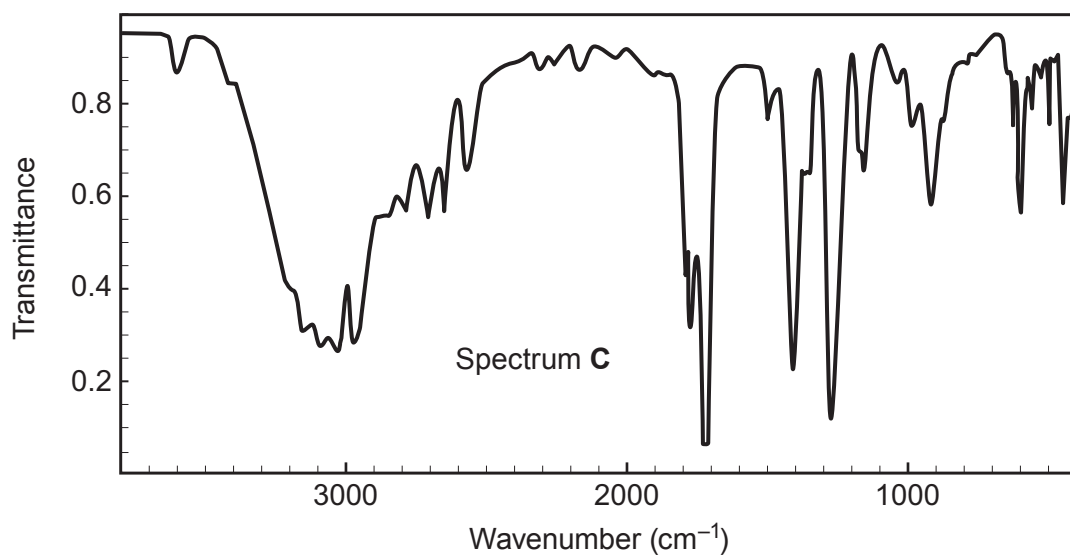
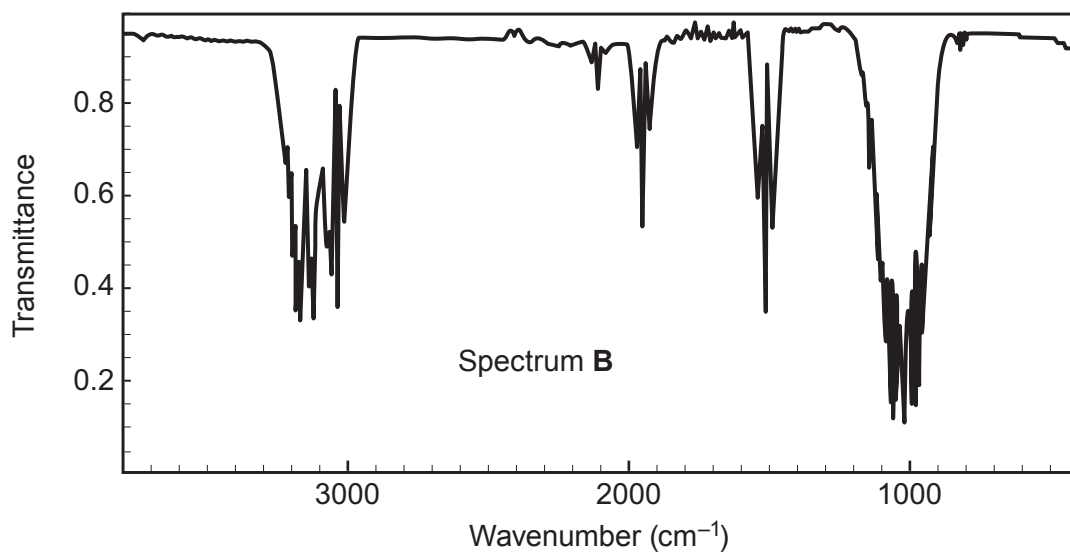
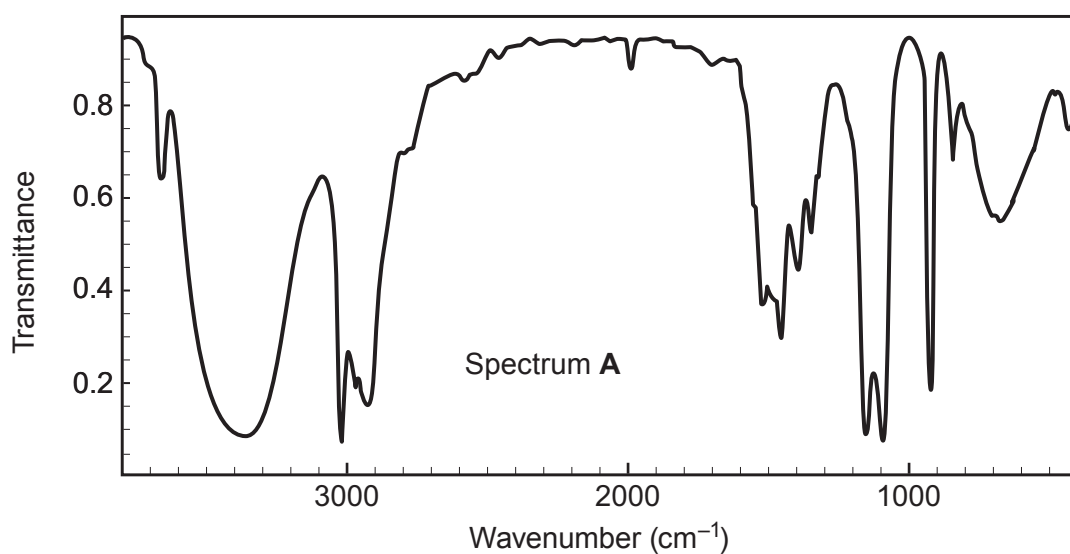
The infrared spectra of ethanol, ethanoic acid and one other compound are shown on the opposite page.

Use the information in the table to identify the spectra belonging to ethanol and ethanoic acid. [2]

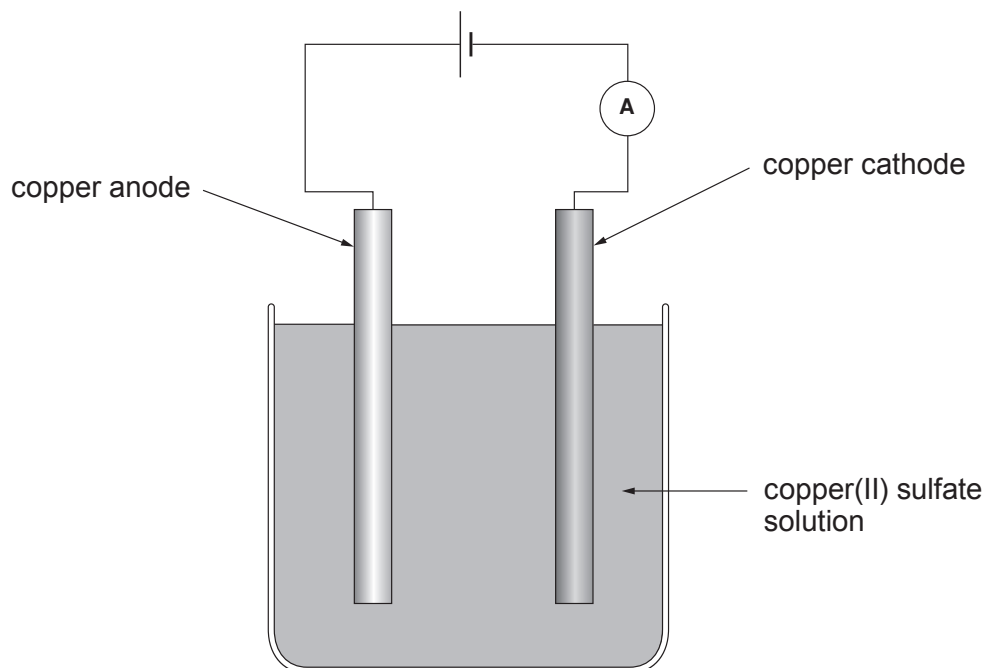
Ethanol

Ethanoic acid





8. A student used the apparatus shown to investigate how changing current affects the mass of copper deposited on the cathode.



Equal volumes and concentrations of copper(II) sulfate solution were used each time. Each experiment was run for 10 minutes. All the readings were obtained at room temperature. The following procedure was followed before the mass of copper could be found.

- the electrode was carefully removed from the electrolyte
- the copper deposit was washed
- the electrode and copper deposit were dried

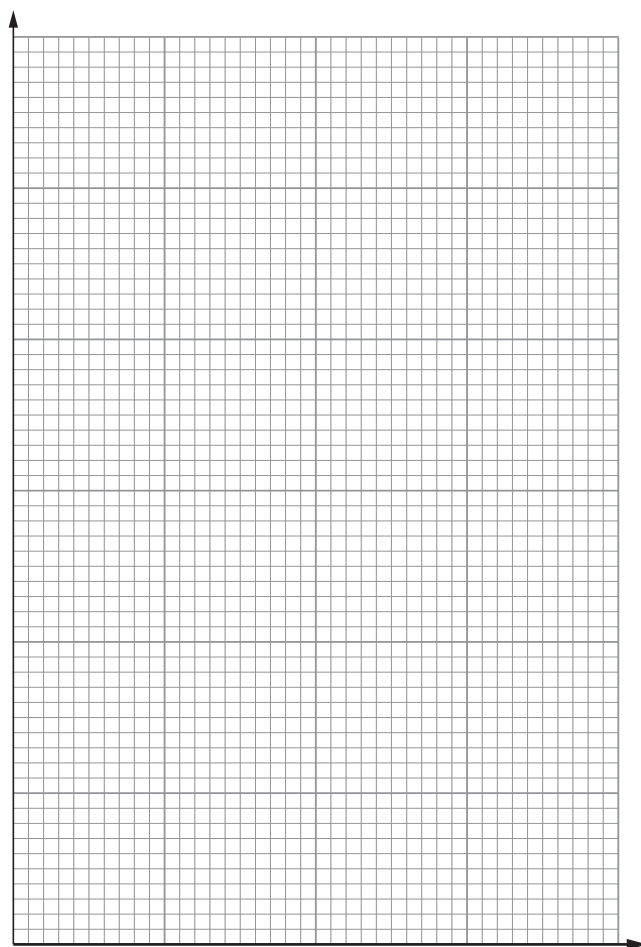
Her results are shown in the following table.

Current (A)	Mass of copper deposited (g)
0.0	0.00
0.5	0.16
1.0	0.30
1.5	0.43
2.0	0.60



- (a) Choose suitable scales for the axes below and plot the current against the mass of copper deposited. Draw a suitable line. [4]

Mass of
copper
deposited
(g)



Current (A)

- (b) Use your graph to predict the mass of copper deposited using a current of 3.5 A. [1]

Mass of copper deposited = g



- (c) Explain why the electrolyte keeps its blue colour during the electrolysis process. Include electrode equations to support your answer. [4]

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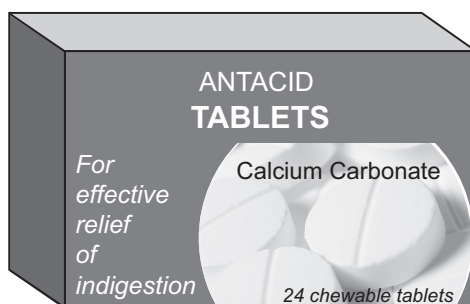
Examiner
only

9



9. Indigestion is caused by excess hydrochloric acid in the stomach. The calcium carbonate in an antacid tablet neutralises the excess acid.

Calcium carbonate reacts with hydrochloric acid according to the equation below.



To determine how much calcium carbonate is present, a crushed indigestion tablet was mixed with water and titrated with dilute hydrochloric acid of concentration 1.0 mol/dm^3 . The end-point was determined using the indicator phenolphthalein. The procedure was repeated three times and the mean volume of hydrochloric acid solution needed to neutralise one tablet was found to be 15.2 cm^3 .

- (a) Calculate the number of moles of hydrochloric acid in 15.2 cm^3 of the 1.0 mol/dm^3 solution. [2]

Number of moles = mol

- (b) Calculate the number of moles of calcium carbonate in one tablet. [1]

Number of moles = mol



- (c) Calculate the mass of calcium carbonate in one tablet. Give your answer in **milligrams**, **mg**. [3]

$$A_r(\text{Ca}) = 40$$

$$A_r(\text{C}) = 12$$

$$A_r(\text{O}) = 16$$

Mass of calcium carbonate = mg

- (d) The diagram shows the labelling on the packet of indigestion tablets.

Each tablet contains:

calcium carbonate	680 mg
magnesium carbonate	80 mg
sucrose and glucose	250 mg
flavouring	
talc	
saccharin	
sodium and magnesium stearate	

Suggest an explanation for the difference between your answer in part (c) and the information given on the packet. [1]

.....

.....

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1																				4 He Helium 2			

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3410UB0-1



S19-3410UB0-1

THURSDAY, 16 MAY 2019 – MORNING**CHEMISTRY – Unit 2:****Chemical Bonding, Application of Chemical Reactions and Organic Chemistry****HIGHER TIER**

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(c) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

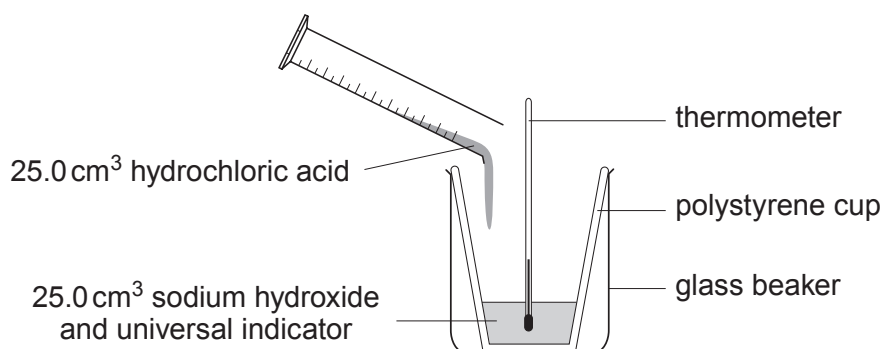
For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	7	
3.	4	
4.	9	
5.	7	
6.	12	
7.	11	
8.	13	
9.	8	
Total	80	

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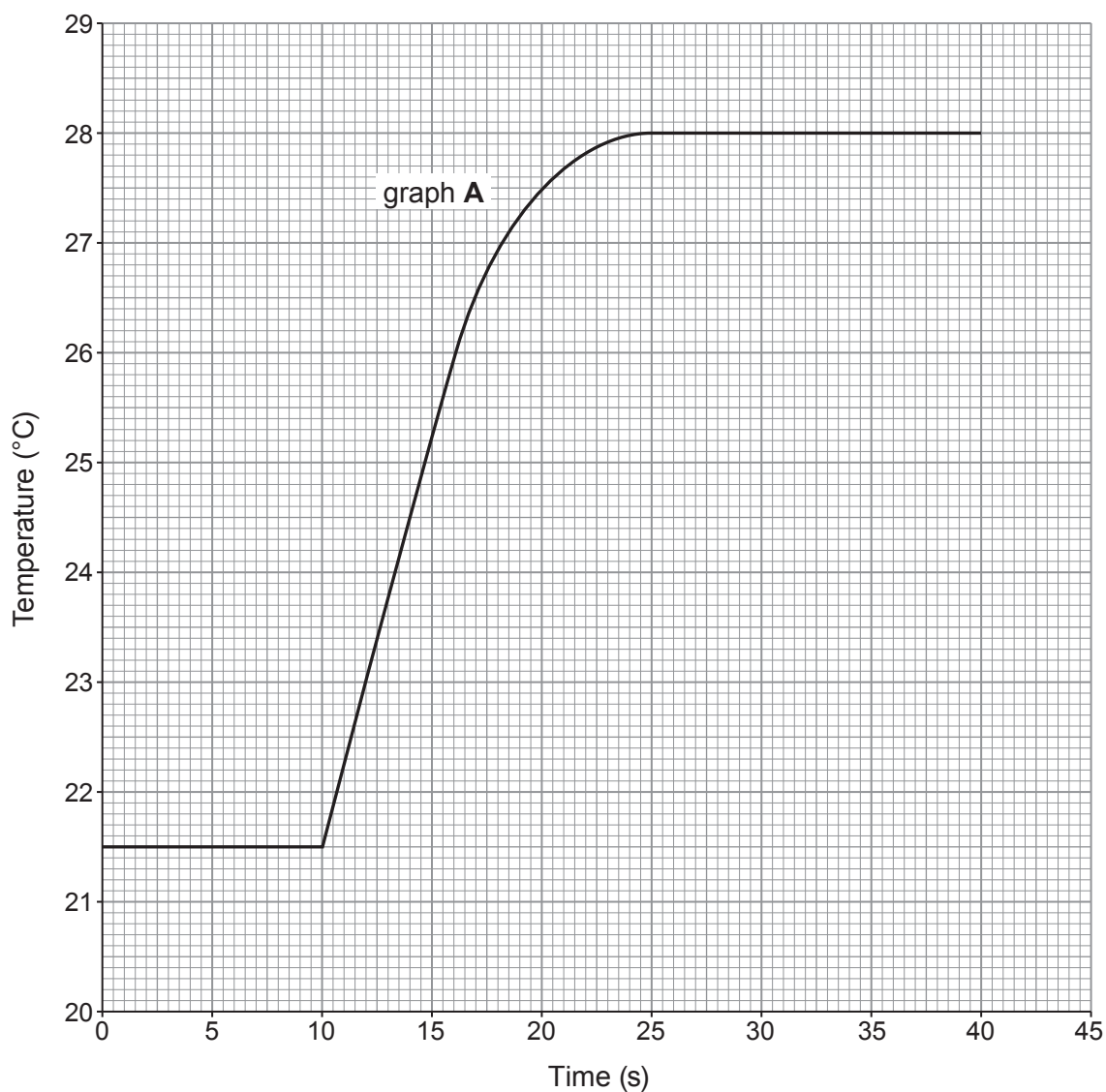
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Answer all questions.

1. A student investigated the temperature rise during a neutralisation reaction.



The student put 25.0 cm³ of sodium hydroxide solution and 5 drops of universal indicator into a polystyrene cup and recorded the temperature of the alkali. After 10 seconds the student added 25.0 cm³ of dilute hydrochloric acid to the alkali and recorded the temperature every 5 seconds for another 30 seconds. Graph **A** shows the results obtained.



- (a) (i) Use the graph to find the maximum temperature rise during the reaction. [1]

Temperature rise = °C

- (ii) The energy given out can be calculated using the following formula.

energy given out = **total** volume of reaction mixture × 4.2 × temperature rise

Calculate the energy given out during the reaction. [2]

Energy given out = J

- (iii) The temperature of the contents in the cup was recorded after 2 hours. Give the final temperature reading you would expect. Give the reason for your answer. [1]

Final temperature °C

Reason

.....

- (b) The student repeated the experiment using 25.0 cm³ of ethanoic acid of the same concentration as the hydrochloric acid. The table shows the results obtained.

Time (s)	0	5	10	15	20	25	30	35	40
Temperature (°C)	21.5	21.5	21.5	24.0	26.0	26.9	27.0	27.0	27.0

Plot the results on the grid on page 2. Draw a suitable line. Label your graph **B**. [2]

- (c) Use the graphs to state which of the two acids is the stronger – hydrochloric acid or ethanoic acid. Give the reason for your choice. [1]

Acid

Reason

.....



(d) The temperature rises in both experiments were much **lower** than expected. The student suggested that using a temperature sensor instead of a thermometer would give temperature rises closer to the expected values.

(i) State why using a temperature sensor would still give a lower than expected temperature rise. [1]

.....

.....

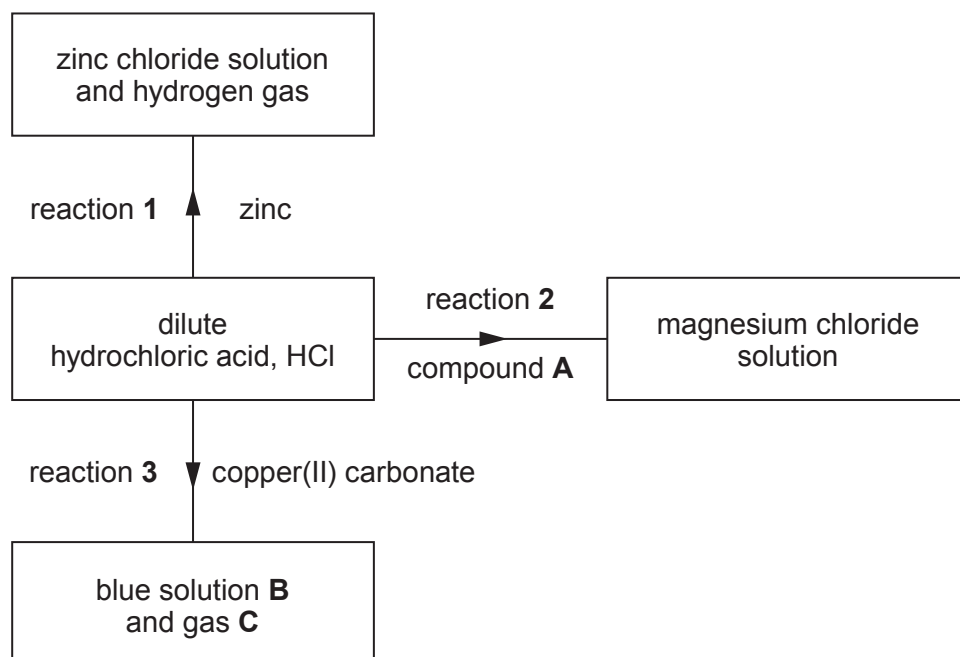
(ii) What improvement to the apparatus would you suggest to the student to obtain temperature rises closer to the expected values? [1]

.....

.....



2. The diagram shows three reactions which are used to prepare soluble salts.



(a) (i) Name compound **A**. [1]

.....

(ii) Give the names of blue solution **B** and gas **C** formed in reaction 3. [2]

blue solution **B**

gas **C**

(b) Write the balanced symbol equation for reaction 1. [2]

.....

(c) Reaction 1 was repeated using magnesium instead of zinc. Explain the difference, if any, that you would expect to see. [2]

.....

.....

.....



3. The tables show the molecular formulae of some alkanes and alkenes.

Alkanes
CH_4
C_2H_6
C_3H_8
C_4H_{10}

Alkenes
C_2H_4
C_3H_6

- (a) The general formula for the alkene family is C_nH_{2n} . Give the general formula for the alkane family. [1]

.....

- (b) When alkanes and alkenes completely burn in air they form the same two products.

Give the chemical formulae for both products. [1]

..... and

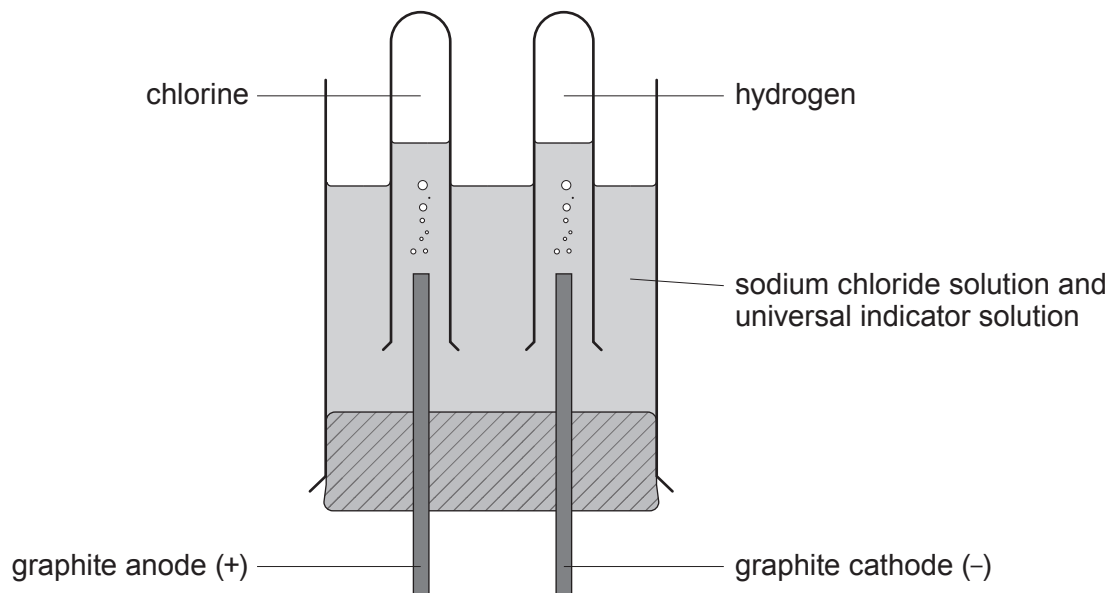
- (c) Draw the structural formula for propene. [1]

- (d) Bromine water is used to distinguish alkenes from alkanes. Describe the colour **change** seen when bromine water is added to an alkene. [1]

.....



4. (a) The diagram shows apparatus that can be used for the electrolysis of sodium chloride solution.



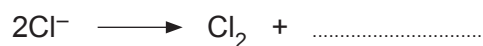
- (i) Explain why hydrogen, and **not** sodium, is formed at the cathode. [1]

.....

.....

.....

- (ii) Complete the electrode equation for the reaction at the anode. [1]



- (iii) Explain why the universal indicator turns purple during electrolysis. [2]

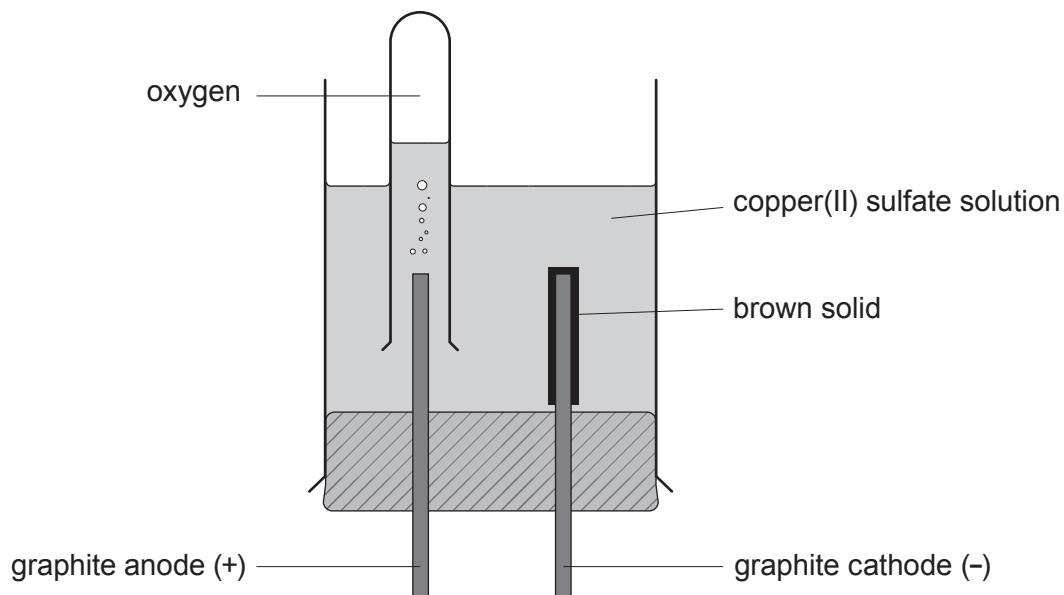
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(b) The diagram shows the electrolysis of copper(II) sulfate solution.



- (i) Explain, using the reaction occurring at the cathode, the meaning of the term *reduction*. [1]

.....

.....

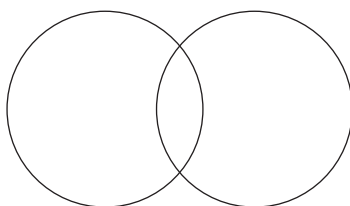
- (ii) Over time the electrolyte turns from blue to colourless. State the change you would make to the apparatus so that the electrolyte remains blue during the process. Give a reason for your answer. [2]

.....

.....

.....

- (iii) The electronic structure of oxygen is (2,6). Complete the diagram showing the outer shell electrons in an oxygen molecule, O_2 . [2]



5. (a) Smart materials are used to make the frames and lenses of certain spectacles.



Give the names of the different types of smart material used. Describe the unusual property of each. [2]

Frames

.....

.....

Lenses

.....

.....



(b) Titanium dioxide has been used in sun creams for decades. Sun screens available today use nano-scale titanium dioxide particles. Some people believe using creams containing nano-particles is unsafe.

(i) Give the advantage of using nano-scale titanium dioxide particles rather than larger titanium dioxide particles to make sun screens. [1]

.....

.....

(ii) Explain why some people are concerned about the use of nano-scale titanium dioxide particles in sun screens. [2]

.....

.....

.....

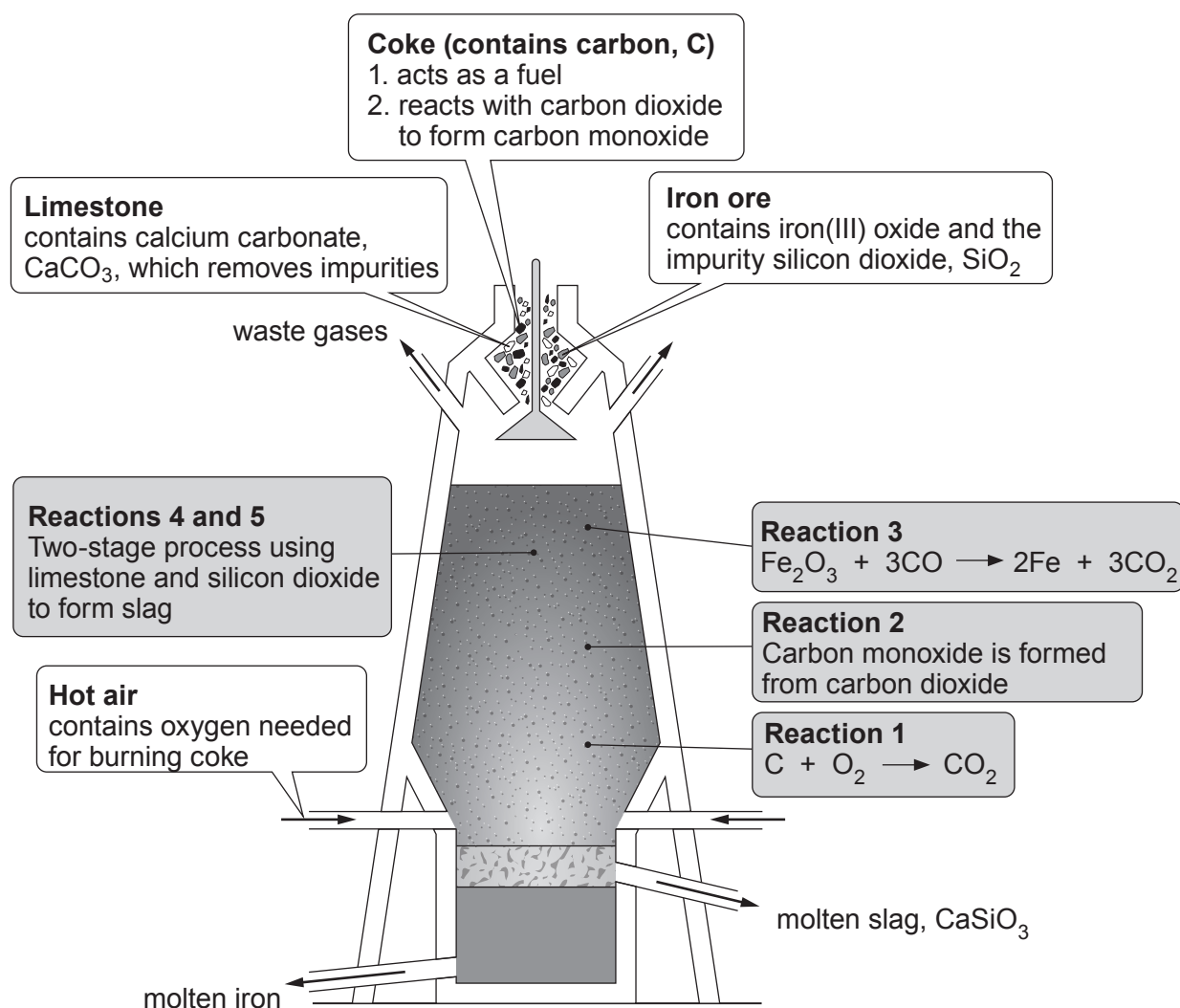
(iii) **Approximately** how many times bigger are common titanium dioxide particles ($3 \times 10^{-7} \text{ m}$) than nano-scale titanium dioxide particles ($2.5 \times 10^{-10} \text{ m}$)? [2]

Answer =

7



6. (a) Iron is extracted from its ore in the blast furnace.



Use information from the diagram and your knowledge to answer parts (i) and (ii).

- (i) Write a balanced symbol equation for reaction 2. [2]

.....

- (ii) Describe the two-stage process to form slag. [3]

.....

.....

.....

.....



(b) Iron(III) oxide reacts with dilute hydrochloric acid forming iron(III) chloride and water.

(i) Balance the symbol equation for this reaction. [1]



(ii) Sodium hydroxide solution can be used to detect the presence of aqueous iron(III) ions.

The symbol equation below represents the reaction occurring between solutions of sodium hydroxide and iron(III) chloride.



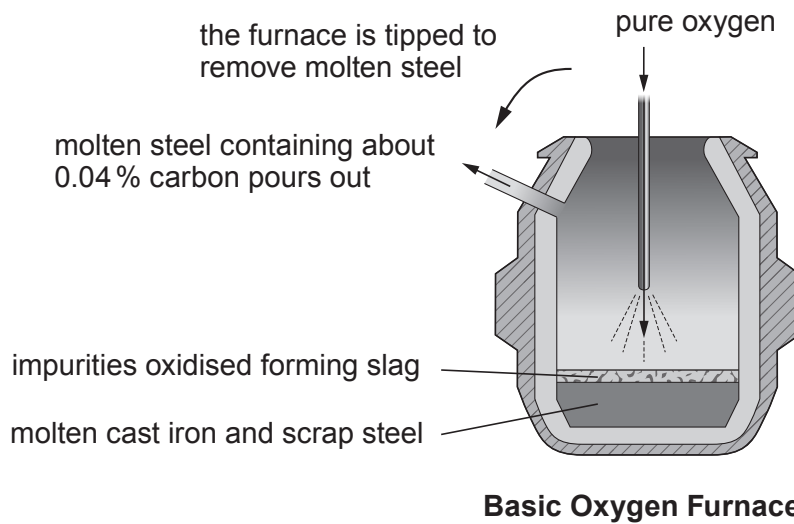
Write the **ionic** equation for the formation of the precipitate. [2]

.....



(c)

Steel manufacture in the UK



One of the major steelmaking processes used today in the UK is the Basic Oxygen Furnace, BOF. The raw materials for the BOF are cast iron from a blast furnace and scrap steel. Oxygen (>99.5% pure) is “blown” into the BOF at supersonic speed. The impurities are oxidised producing great quantities of heat which melts the scrap steel.

Steel can be described in general terms as iron containing small amounts of carbon, to make it tougher and more ductile. There are many types of steel, each with its own specific chemical composition and properties to meet the needs of the many different applications. **Figure 1** below shows the relationships between the carbon content of steel and its ductility, tensile strength and hardness.

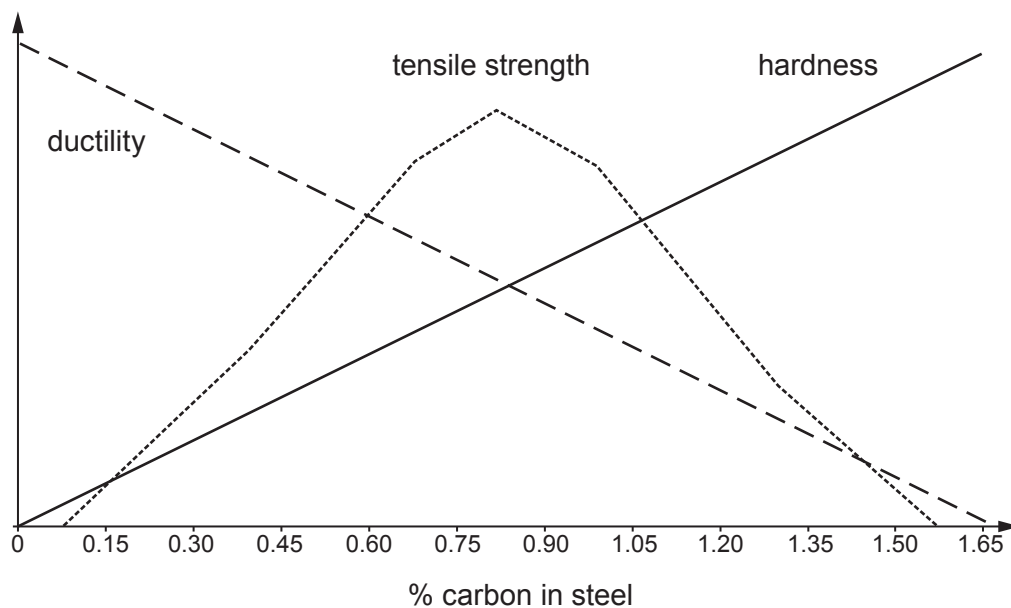


Figure 1

Ductility is a material's ability to be pulled into a wire.

Tensile strength is a measurement of the force required to pull something such as wire to the point where it breaks.

Hardness is a measure of how resistant a material is to permanent shape change when a compressive force is applied.



Figure 2 shows the percentage of carbon in various alloys of iron.

Name of alloy	Percentage of carbon
low carbon steel	0.0 – 0.6 %
medium carbon steel	0.6 – 0.8 %
high carbon steel	0.8 – 1.3 %
very high carbon steel	1.3 – 1.6 %

Figure 2

- (i) Tick (✓) the box next to the statement which best describes **one** way that production costs are reduced. [1]

high purity oxygen is used

☐

impurities are oxidised forming heat

☐

oxygen is blasted in at supersonic speed

☐

scrap steel is used in the process

☐

- (ii) Tick (✓) the box next to the statement which best describes the effect of increasing the percentage of carbon in steel from 0 % to 0.45 %. [1]

ductility increases, hardness increases

☐

tensile strength increases, ductility increases

☐

ductility decreases, tensile strength increases

☐

hardness increases, tensile strength decreases

☐

- (iii) A steel manufacturer wants to design an alloy with a high tensile strength but low ductility. Tick (✓) the box next to the approximate value for the percentage of carbon that should be included. [1]

0.2

☐

0.6

☐

1.0

☐

1.5

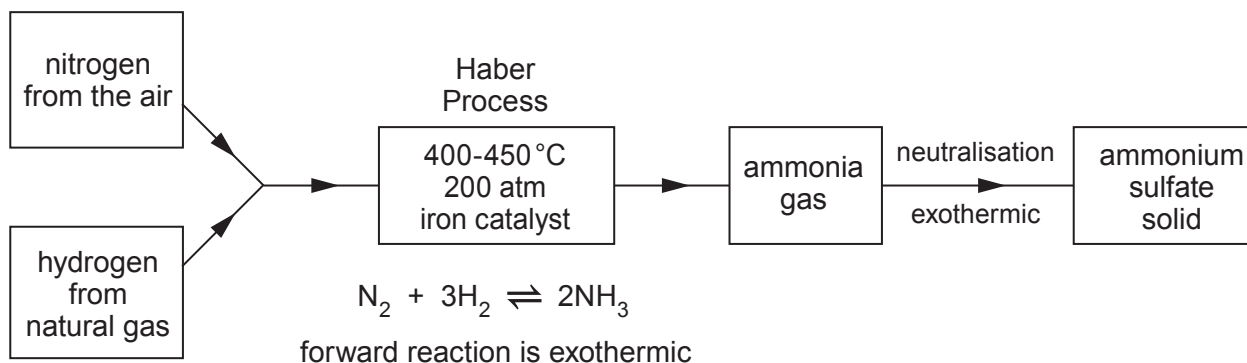
☐

- (iv) Name the alloy which is the most easily pulled into a wire and withstands the least compressive force. [1]

.....



7. The flow diagram outlines the manufacture of ammonia by the Haber Process followed by the production of ammonium sulfate by neutralisation.



- (a) Explain the choice of temperature used in the Haber Process and the reason why a catalyst is used. [3]

.....

.....

.....

.....

- (b) Powdered ammonium sulfate is industrially formed by spraying sulfuric acid into a reaction chamber filled with ammonia gas. The exothermic reaction that occurs evaporates all water present in the system, forming a powdery salt.

Write a balanced symbol equation for this reaction. [2]

.....



8. Methanol, ethanol and butanol all belong to the homologous series of alcohols.

(a) The table shows information about two different methods for the manufacture of ethanol.

	Method A fermentation	Method B addition reaction
Raw material	sugar from sugar cane	ethene from crude oil
Reaction	yeast catalyst $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$	phosphoric(V) acid catalyst $C_2H_4(g) + H_2O(g) \rightarrow C_2H_5OH(l)$
Operating pressure	1 atm	60 atm
Type of process	batch (stop-start)	continuous (runs all the time)

Use the information in the table and your knowledge to answer the following question.

Explain **two** advantages and **two** disadvantages of method **A** compared with method **B**.
[4]

Advantages

1.

.....

2.

.....

Disadvantages

1.

.....

2.

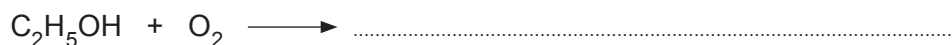
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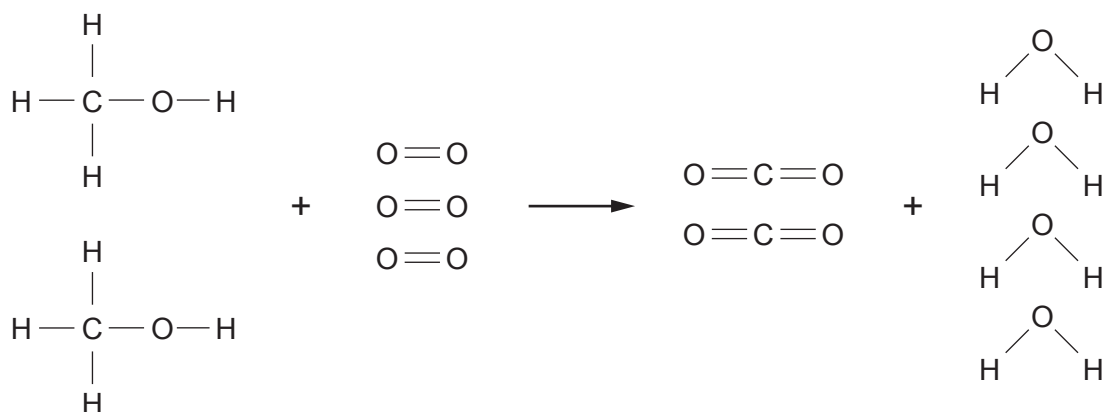
- (b) Ethanol is found in alcoholic drinks. Alcoholic drinks turn sour when left exposed to air because ethanol is oxidised to ethanoic acid and water.

Complete the balanced symbol equation for this reaction.

[1]



- (c) Methanol readily burns in air. The diagram shows the bonds which are broken and the bonds which are formed during the combustion of methanol.



Some relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
C—H	413
C—O	358
C=O	805
O—H	464



- (i) The total energy needed to break all the bonds in the reactants is 5616 kJ. The energy needed to break the bonds in one molecule of methanol is 2061 kJ.

Use this information to calculate the amount of energy needed to break **one** $\text{O}=\text{O}$ bond. [2]

Energy needed = kJ

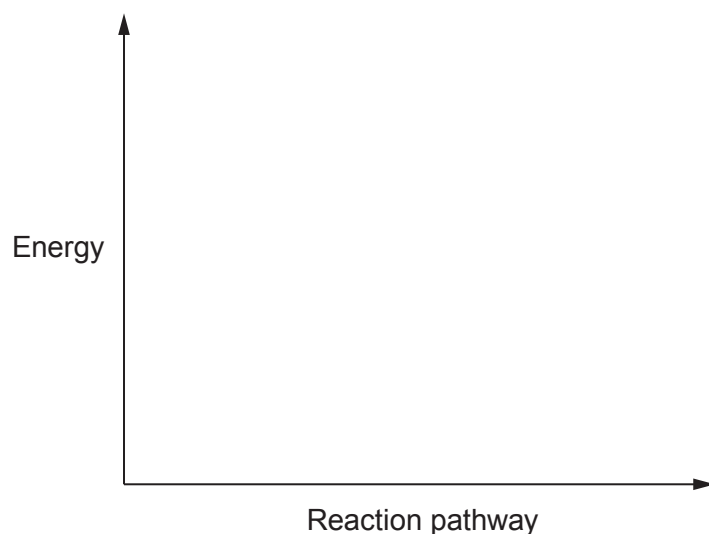
- (ii) Calculate the total energy released when all the bonds in the products are formed. [2]

Energy released = kJ

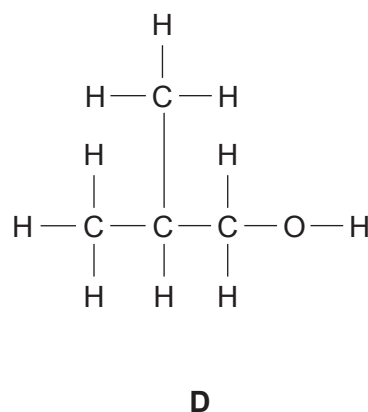
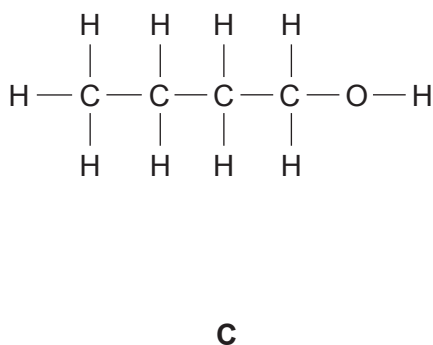
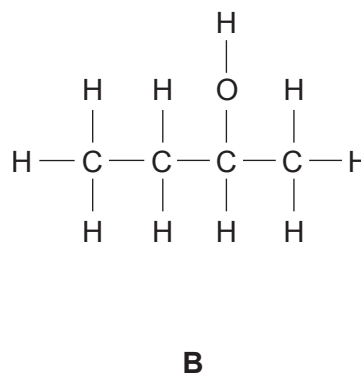
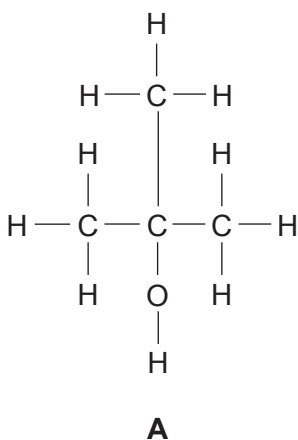
- (iii) The burning of methanol gives out heat and is said to be exothermic. Use the total energy value 5616 kJ and your answer to part (ii) to show that this is correct. [1]

.....
.....

- (iv) On the axes below draw the energy profile for the combustion of methanol and use the symbol ($\updownarrow$) to show the activation energy for the reaction. [1]



(d) Butanol has the molecular formula C_4H_9OH . It has four positional isomers, **A**, **B**, **C** and **D**.



Give the **letter** of the isomer corresponding to each of the names in the table.

[2]

Name	Isomer
butan-1-ol	
butan-2-ol	
2-methylpropan-1-ol	
2-methylpropan-2-ol	



9. (a) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

	A	B	C
Add dilute HCl	no reaction	fizzes	no reaction
Add $\text{BaCl}_2(\text{aq})$	white precipitate forms	no reaction	no reaction
Add $\text{NaOH}(\text{aq})$	green precipitate forms	pungent smelling gas given off, turns damp red litmus paper blue	no reaction
Add $\text{AgNO}_3(\text{aq})$	no reaction	no reaction	cream precipitate forms
Flame test	no colour	no colour	apple-green flame

Use the information provided to give the **chemical name** for each of the compounds. [3]

Compound **A**

Compound **B**

Compound **C**



(b) A technician wants to prepare 250 cm^3 of a 0.25 mol/dm^3 solution of lead nitrate, $\text{Pb}(\text{NO}_3)_2$.

(i) Calculate the number of moles of lead nitrate required to make the solution. [2]

Number of moles = mol

(ii) Calculate the mass of solid lead nitrate that should be dissolved to make the solution. [2]

$A_r(\text{Pb}) = 207$

$A_r(\text{O}) = 16$

$A_r(\text{N}) = 14$

Mass = g

(iii) The only electronic balance available to the technician has a precision of $\pm 0.01\text{ g}$.

Exactly how much lead nitrate should the technician weigh out to ensure that the concentration of the solution is as close as possible to 0.25 mol/dm^3 ? [1]

Mass g

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

4

2



4

5

6

1

O

1 H Hydrogen 1																				
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10	4 He Helium 2		
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18			
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36			
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54			
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86			
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																		
Key																				

Key

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410UB0-1



Z22-3410UB0-1

FRIDAY, 27 MAY 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **11** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	7	
3.	7	
4.	5	
5.	9	
6.	10	
7.	5	
8.	9	
9.	7	
10.	9	
11.	6	
Total	80	

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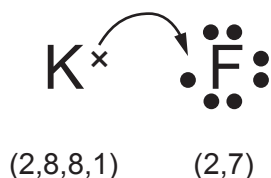
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Answer **all** questions.

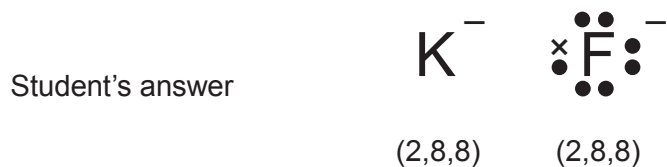
1. (a) The table shows the electronic structure of the elements present in potassium fluoride.

Element	Electronic structure
potassium	2,8,8,1
fluorine	2,7

The diagram shows the electron transfer that occurs when potassium reacts with fluorine to form potassium fluoride. The ● and × symbols are outer shell electrons.



A student was asked to draw a diagram showing the electronic structures and charges on the ions formed. There are **two** mistakes in the student's answer.

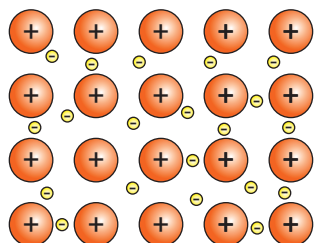


- (i) Circle the **two** mistakes in the student's answer. [2]
- (ii) Name the type of bonding found in potassium fluoride. [1]

.....



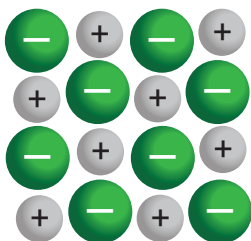
- (iii) The diagrams show four different structures. Give the **letter** of the structure most likely to represent potassium fluoride. [1]



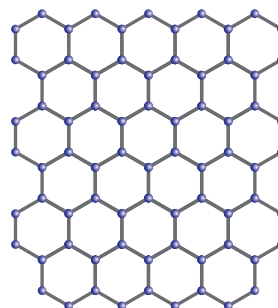
A



B



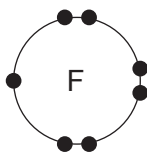
C



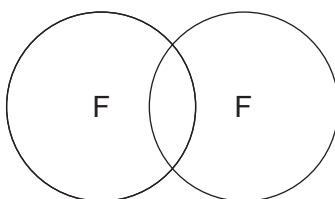
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Letter

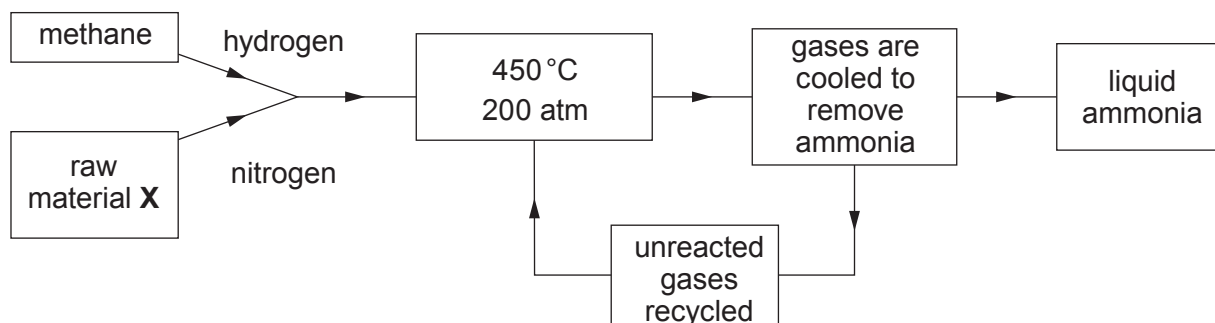
- (b) The diagram shows the electrons in the outer shell of an atom of fluorine.



Complete the diagram to show the outer shell electrons in a molecule of fluorine. [2]



2. (a) The diagram outlines the manufacture of ammonia by the Haber process.



- (i) Name the raw material **X**. [1]
- (ii) The pressure used in the Haber process is 200 atm. State why a higher pressure is **not** used. [1]
-
- (iii) At 450 °C, the reaction is very slow. Iron is used in the process to speed up the reaction. Give the name for a substance used to speed up a chemical reaction. [1]
-
- (iv) The reaction between nitrogen and hydrogen is represented by the equation below.



Complete the equation below using the key:

[2]



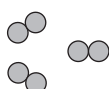
nitrogen gas, N₂



hydrogen gas, H₂



+



(b) One of the main uses of ammonia is in the manufacture of fertilisers.

The table shows the results obtained when tests were carried out on three different fertilisers **A**, **B** and **C**.

Fertiliser	Test for positive ion	Test for negative ion
A	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding barium chloride solution a white precipitate forms
B	Lilac flame test	On adding silver nitrate solution a white precipitate forms
C	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding silver nitrate solution a white precipitate forms

Give the **letter** of the fertiliser which is ammonium sulfate.

[1]

Letter

(c) Ammonia reacts with chlorine to form nitrogen and hydrogen chloride.

Complete the balancing of the equation for this reaction.

[1]



3. (a) The list below shows part of the reactivity series.

sodium
aluminium
(carbon)
tin
copper
silver

- (i) Tin is extracted from its ore by heating with carbon. Aluminium is extracted from its ore using a different method. Give the name of the method used to extract aluminium. [1]

.....

- (ii) The equation shows the extraction of tin from tin oxide using carbon.



Tick (✓) the box next to the correct statement. [1]

Carbon is reduced

☐

Tin is oxidised

☐

Tin oxide is reduced

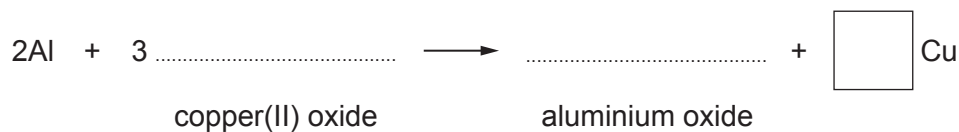
☐

Carbon dioxide is oxidised

☐

- (iii) When aluminium and copper(II) oxide are heated together, aluminium oxide and copper are formed.

Complete and balance the equation for this reaction. [3]



- (b) A teacher wanted to find out the position of four metals **A**, **B**, **C** and **D** in the reactivity series.

She heated each metal in turn with oxides of the other three. The results were as follows.

A reduced the oxide of **C**

B reduced the oxide of **A**

B reduced the oxide of **C**

D reduced the oxide of **B**

Place the metals in order of reactivity.

[2]

Most reactive

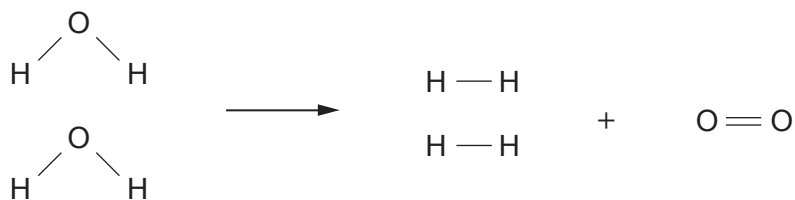
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.....

Least reactive



4. The equation shows the bonds which are broken and the bonds which are formed during the electrolysis of water.



- (a) The total energy needed to break the bonds in the reactant is 1856 kJ.

Calculate the energy needed to break **one** O — H bond.

[2]

Energy = kJ

- (b) The total energy released when the bonds in the products are formed is 1370 kJ.

The bond energy for H — H is 436 kJ.

Calculate the energy released when making **one** O = O bond.

[2]

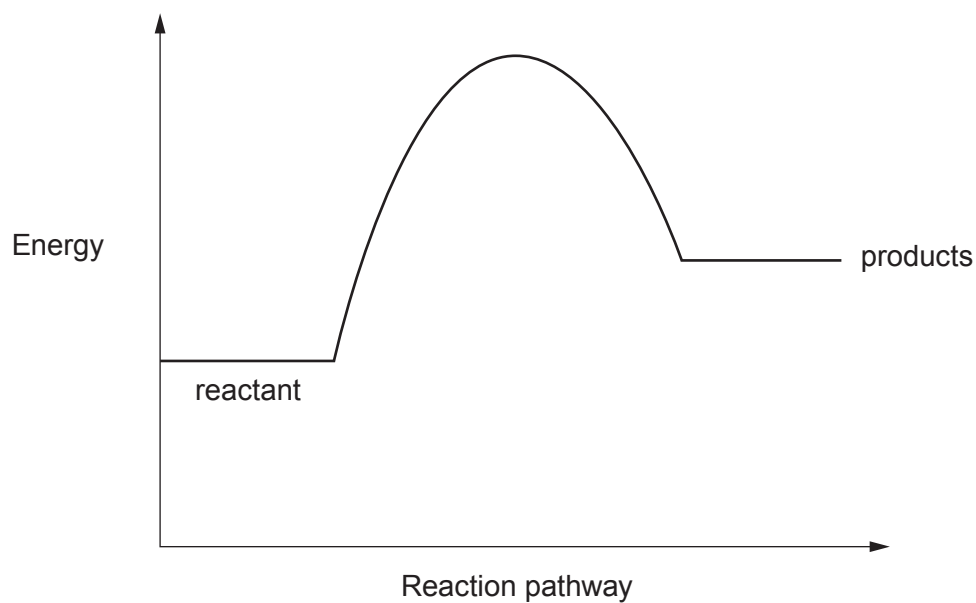
Energy = kJ



- (c) The energy profile diagram shows the reaction to be endothermic.

[1]

On the diagram below use the symbol ($\updownarrow$) to show the overall energy change for the reaction.

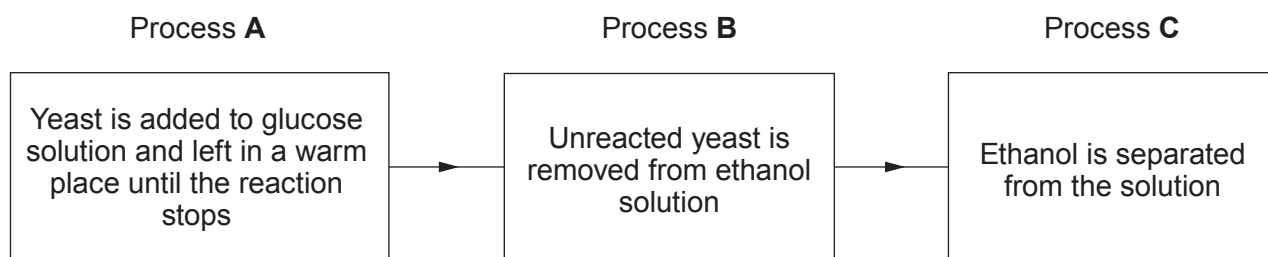


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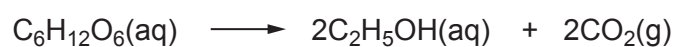
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5. The chart describes the laboratory preparation of ethanol from glucose solution.



The equation for the reaction occurring in process **A** is shown below.



(a) (i) Give the name of the processes **A**, **B** and **C**.

[3]

A

B

C



(ii) A teacher wanted to show that ethanol is collected in process **C**.

- I. Tick (✓) the box next to the chemical test that the teacher would carry out to positively identify the liquid as ethanol. [1]

add bromine water

☐

add acidified potassium dichromate solution

☐

add silver nitrate solution

☐

add barium chloride solution

☐

- II. Tick (✓) the box next to the observation you would expect. [1]

orange to colourless

☐

orange to green

☐

green to orange

☐

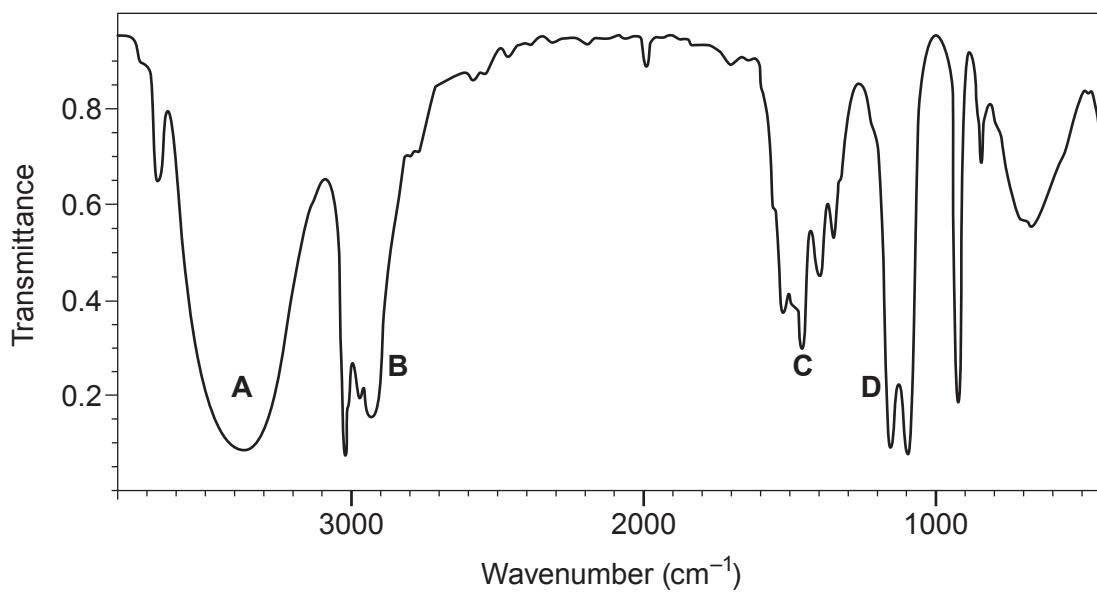
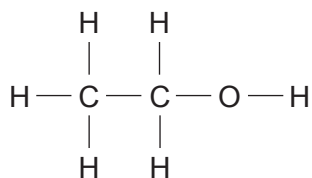
colourless to green

☐

(b) Alcohols can also be identified using infrared spectroscopy.

Bond	Wavenumber (cm^{-1})
$\text{C}=\text{C}$	1620 to 1670
$\text{C}=\text{O}$	1650 to 1750
$\text{C}-\text{H}$	2800 to 3100
$\text{O}-\text{H}$	2500 to 3550

(i) The structural formula and the infrared spectrum of ethanol are shown.



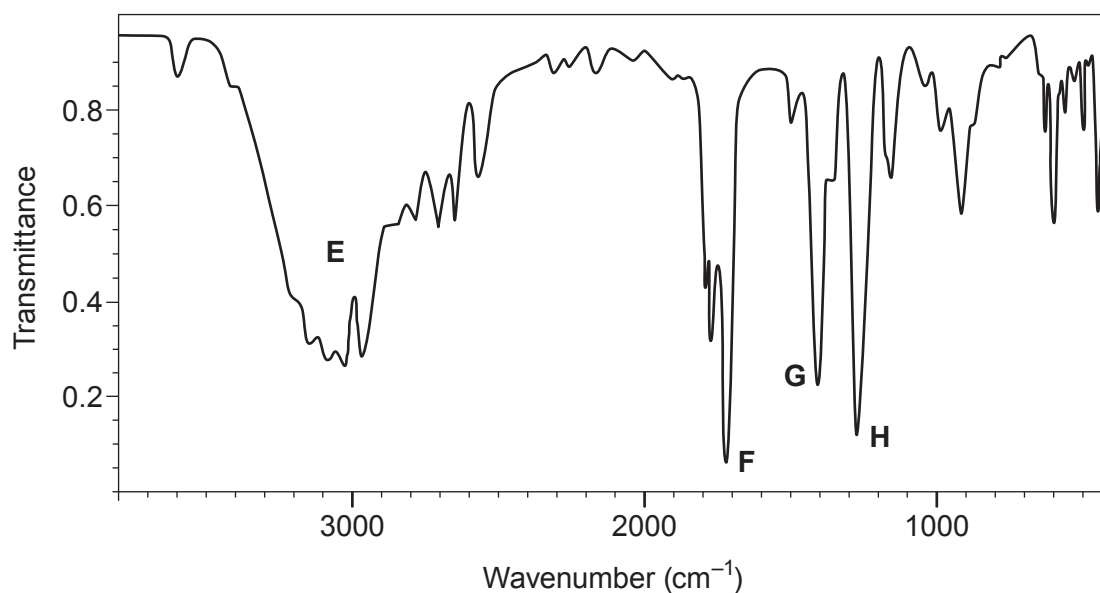
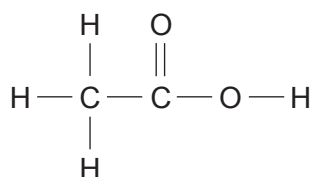
Give the **letter** of the peak that can be used to identify an alcohol.

[1]

Letter



(ii) The structural formula and the infrared spectrum of ethanoic acid are shown.



Give the **letter** of the peak that can be used to distinguish ethanoic acid from ethanol.

[1]

Letter

(c) Ethanol is found in alcoholic drinks.

(i) Give **one** health problem associated with alcohol abuse over a **long** period of time.

[1]

.....

(ii) Give **one** social problem associated with the excessive intake of alcohol (binge drinking) during an evening.

[1]

.....



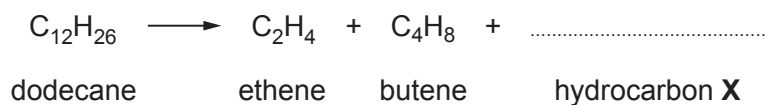
6. (a) Kerosene is one of the fractions separated during the fractional distillation of crude oil.

Kerosene contains dodecane, $C_{12}H_{26}$. Dodecane undergoes a further process **A** to form smaller, more useful hydrocarbons, ethene, butene and one molecule of hydrocarbon **X**.

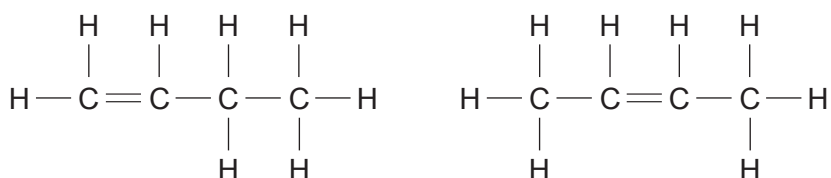
- (i) Name process **A**. [1]

.....

- (ii) Complete the equation for this reaction. [1]



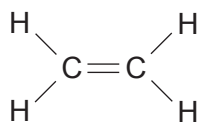
- (b) C_4H_8 has three isomers. The diagrams below show two of the three isomers.



The third isomer is methylpropene. Draw its structure. [1]



- (c) (i) Polyethene is made from ethene by addition polymerisation. Draw the structure of the repeating unit for polyethene. [1]

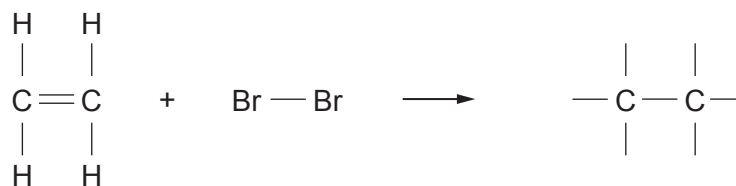


ethene

repeating unit for polyethene

- (ii) The presence of the double bond in ethene can be confirmed using bromine water.

- I. Complete the equation for the reaction between ethene and bromine. [1]



- II. Tick (✓) the box next to the name of the product formed. [1]

1,2-dibromoethene

☐

1,1-dibromoethane

☐

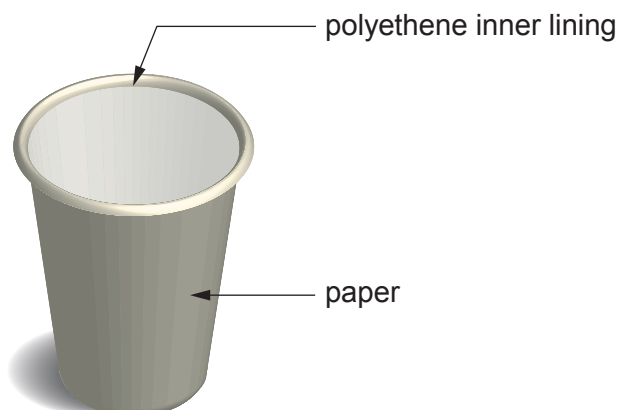
1,2-dibromoethane

☐

1,1-dibromoethene

☐

- (d) In the UK, around 3 billion disposable coffee cups are used every year. Each paper coffee cup is lined with polyethene which is impossible to remove at a recycling plant. If the automated machine that sorts waste at a recycling centre detects a plastic lining, it rejects the cup sending it to general household waste.



- (i) Most general household waste is disposed of in landfill sites.

Apart from using landfill sites, give the **other** disposal method of general household waste. State an environmental problem associated with the method. [2]

Method

Problem

.....

- (ii) Recycling helps to conserve raw materials. Name the raw material used to make polyethene. Give the main reason why it is important to conserve this raw material. [2]

Raw material

Main reason

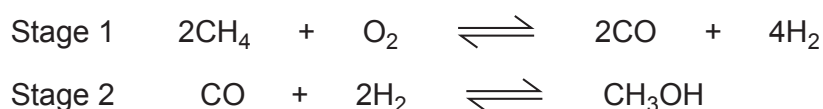
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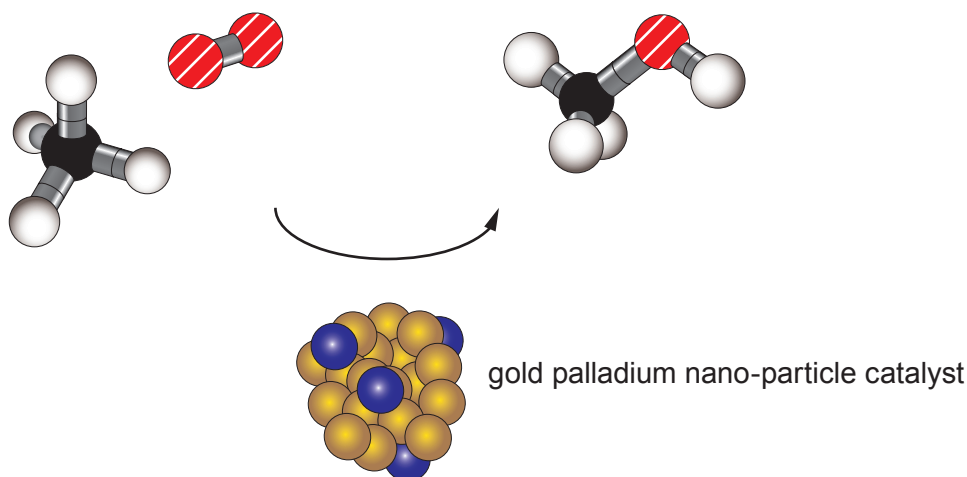
7. Scientists make 'new fuel' discovery

Methanol is vital in manufacturing a wide range of fuels and chemicals. Methanol is produced from methane found in natural gas. At the moment methane is condensed into liquid natural gas at the site where it is extracted and transported in pressurised containers.

Traditionally, methanol is created by converting methane into hydrogen and carbon monoxide molecules at high temperature, then rearranging the atoms in a different order in a second highly pressurised process. The current two-stage process is very energy intensive, as it burns a lot of fossil fuel to achieve high temperatures.



Scientists have discovered a new way of creating greener and cheaper methanol from methane using gold palladium nano-particles to initiate a one-stage chemical reaction that can be done at temperatures no higher than 50 °C. In this new process the gold palladium nano-particles act as a catalyst enabling methane and oxygen to combine, forming methanol in a single stage reaction.



The discovery opens up the prospect of easily converting methane into methanol at the site where the methane is extracted, so that methanol can be transported as a liquid at atmospheric pressure.

Nano-particles have very different and unique properties from their bulk form.

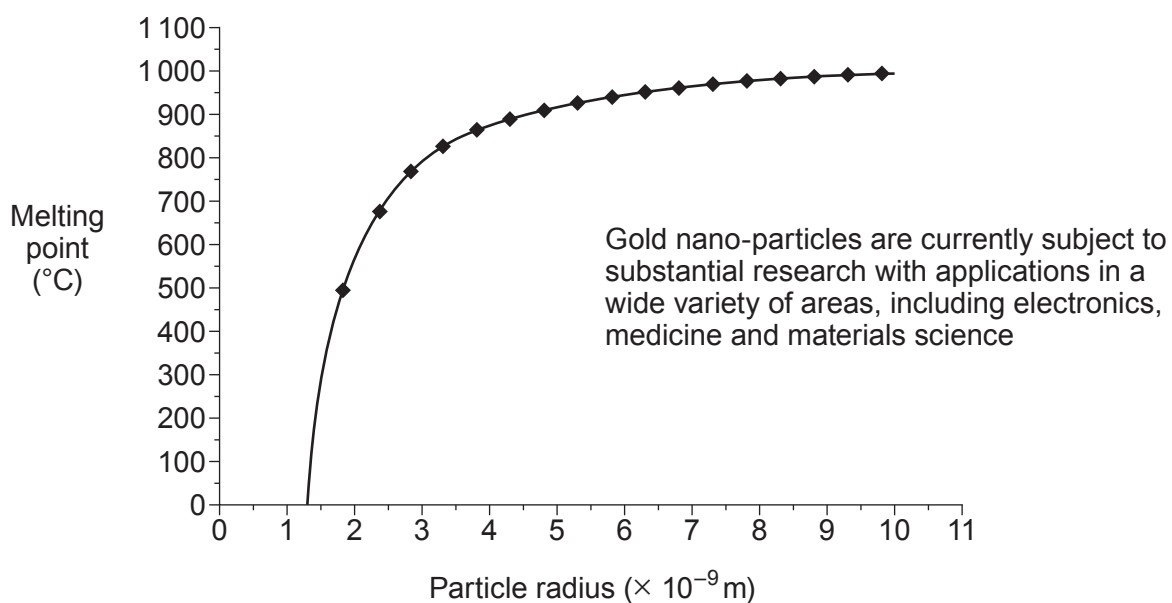
Bulk gold

- shiny
- always gold colour
- inert
- conducts electricity
- melting point 1064 °C

Gold nano-particles

- found in a range of colours (100 nm purple; 20 nm red; 1 nm yellow)
- never gold colour
- very good catalysts
- semi-conductors
- range of melting points





- (a) Tick (✓) the box next to the **two** statements which support the opinion that the new method is more **environmentally friendly**. [2]

It is cheaper than the traditional method

☐

It uses less energy

☐

It reduces carbon dioxide emissions

☐

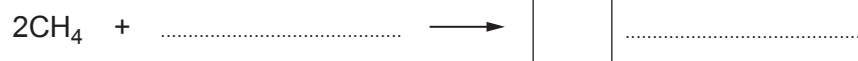
It uses gold nano-particles

☐

It uses more fuel

☐

- (b) Complete the balanced equation for the **new** method of converting methane to methanol. [2]



(c) Tick (✓) the box next to the **true** statement.

[1]

The melting points of gold nano-particles and bulk gold are the same

☐

Gold nano-particles have a fixed melting point value

☐

Smaller gold nano-particles have higher melting points than larger gold nano-particles

☐

The melting point of gold nano-particles depends on their size

☐
☐

5

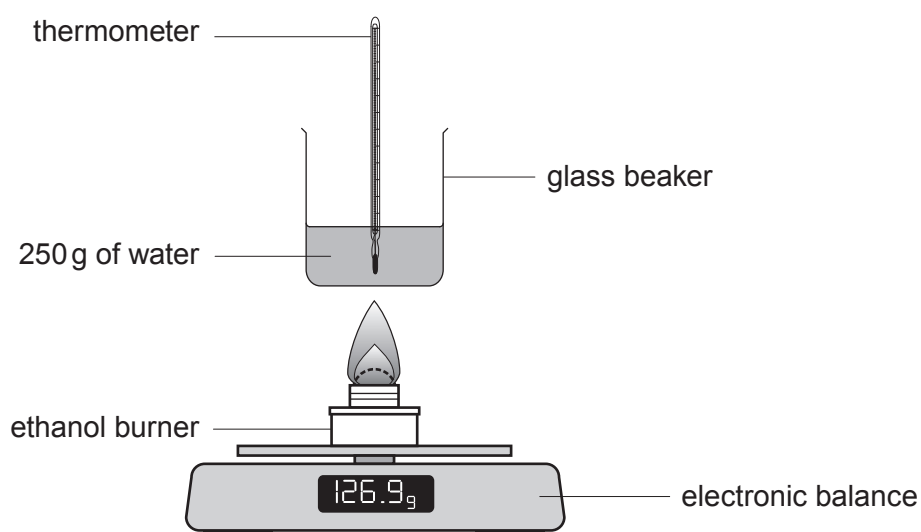


8. Mountaineers often choose an ethanol burner when hiking in extremely cold conditions.



ethanol burner

A group of students was asked to investigate how the mass of ethanol burned is related to the amount of energy given out.



Four students each burned a different mass of ethanol and recorded the temperature rise of 250 g of water. The diagram shows the apparatus used by each of the students. They used their results to calculate the energy given out.

The table shows the students' results.

Student	Mass of ethanol burned (g)	Energy given out ($\text{J} \times 10^4$)
A	1.1	2.2
B	1.8	3.6
C	2.9	5.8
D	4.2	8.4



- (a) The temperature rise can be calculated using the formula below.

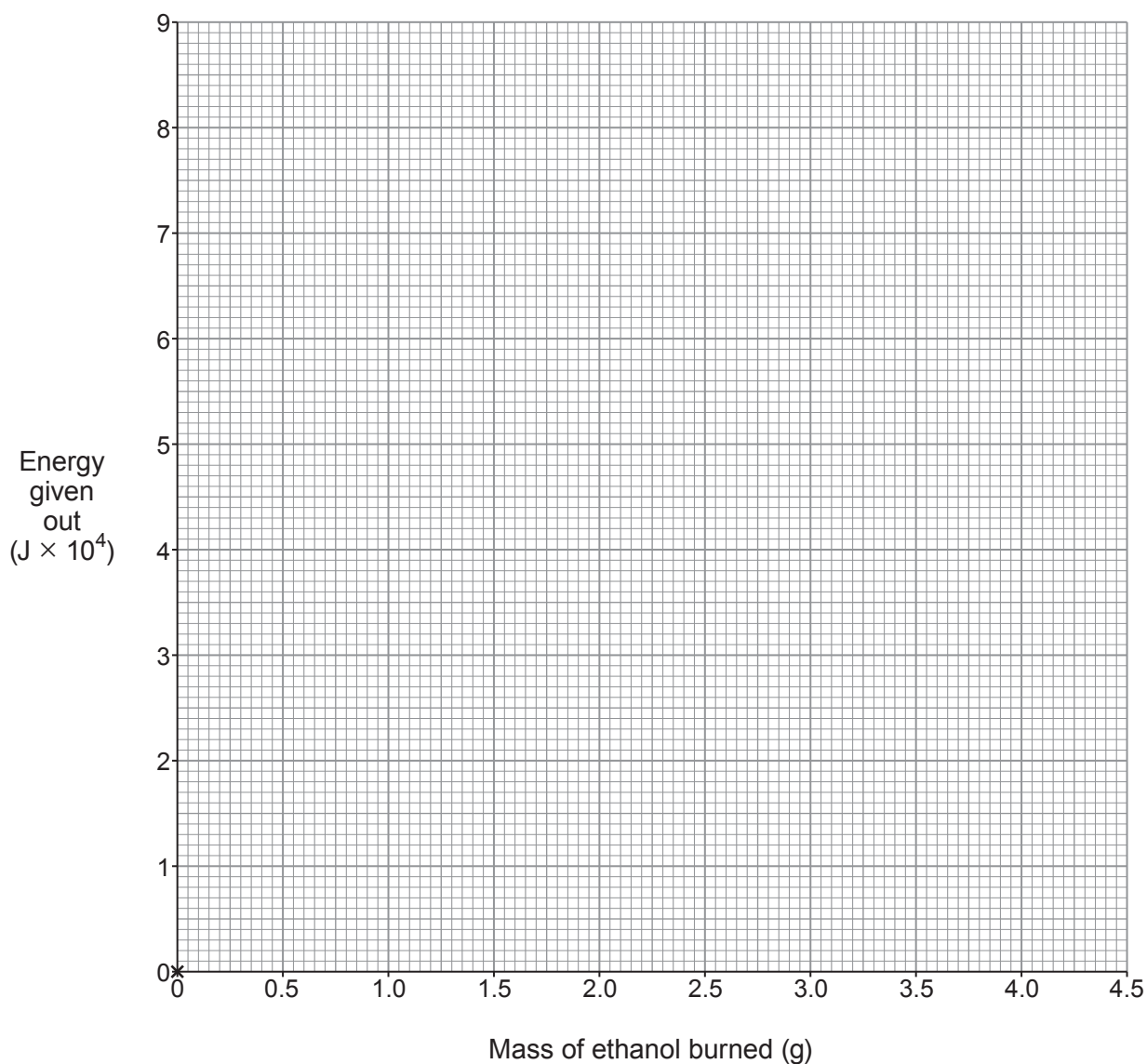
$$\text{energy given out (J)} = 4.2 \times \text{temperature rise (}^{\circ}\text{C)} \times \text{mass of water (g)}$$

Use this formula to calculate the temperature rise recorded by student **D**.

[2]

Temperature rise = $^{\circ}\text{C}$

- (b) Plot the energy given out against mass of ethanol burned on the grid below and draw a suitable line. [3]



- (c) Describe the relationship between the mass of ethanol burned and the energy given out. [2]

.....

.....

.....

- (d) All the recorded temperature rises were **lower** than expected.

Explain **one** piece of advice you would give the students to resolve this problem. [2]

.....

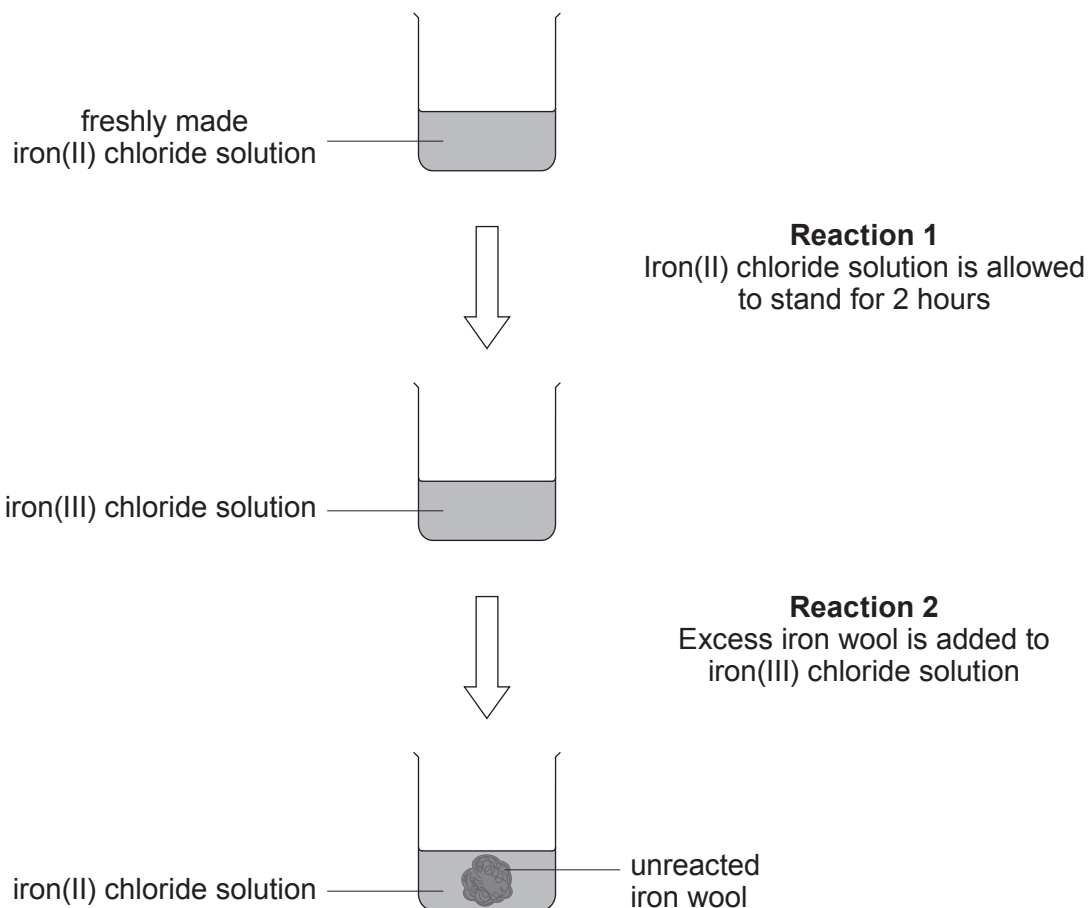
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9. Transition metals have the ability to form ions with different charges. Iron and copper are transition metals.

- (a) A teacher carried out a series of reactions to show that iron can form Fe^{2+} and Fe^{3+} ions.

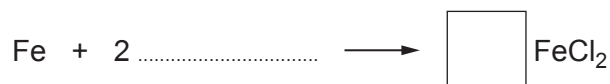


- (i) State what you would expect to **see** during the reactions that shows iron is a transition metal. [1]

.....

- (ii) In **reaction 2**, iron reacts with iron(III) chloride forming iron(II) chloride, FeCl₂.

- I. Complete and balance the equation for this reaction. [2]



- II. Explain the meaning of the term *oxidation* in relation to **reaction 2**. [2]

.....

.....

.....

- (b) Name the reagent used to identify copper(II) ions in solution. Give the observation expected. [2]

Reagent

Observation



10. (a) Some household cleaners are a concentrated solution of ammonia.

To determine the concentration of an ammonia solution, 10.0 cm^3 of a household cleaner was titrated with dilute sulfuric acid of concentration 1.5 mol/dm^3 .

The end-point was determined by using the indicator methyl orange.

The procedure was repeated three times and the mean volume of dilute sulfuric acid needed to neutralise 10.0 cm^3 of the ammonia solution was found to be 12.0 cm^3 .

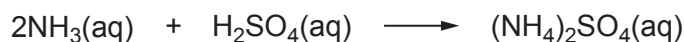


- (i) Use the equation below to calculate the number of moles of sulfuric acid in 12.0 cm^3 of the 1.5 mol/dm^3 solution. [2]

$$\text{concentration} = \frac{\text{number of moles}}{\text{volume}}$$

Number of moles of sulfuric acid = mol

- (ii) Ammonia solution reacts with sulfuric acid according to the equation below.



Use the equation for the reaction to find the number of moles of ammonia in 10.0 cm^3 of the household cleaner. [1]

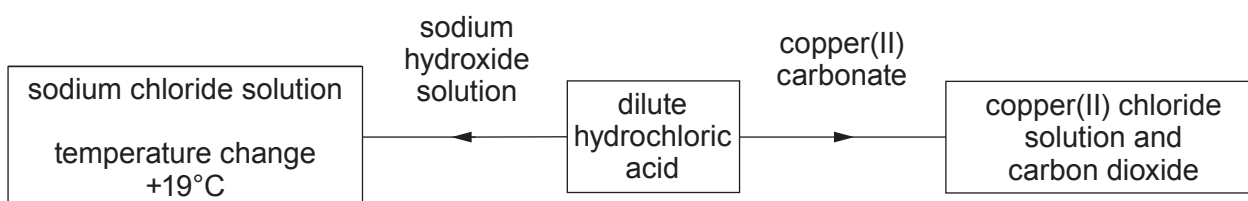
Number of moles of ammonia = mol

- (iii) Calculate the concentration of ammonia in mol/dm^3 . [2]

Concentration of ammonia = mol/dm^3



(b) The diagram shows two reactions of dilute hydrochloric acid.



- (i) Predict the temperature change if dilute ethanoic acid, CH_3COOH , is added to sodium hydroxide instead of dilute hydrochloric acid. Give the reason for your answer. [2]

.....

.....

.....

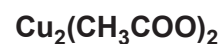
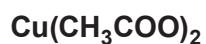
- (ii) A salt is formed when ethanoic acid is added to copper(II) carbonate.

I. Give the name of the salt.

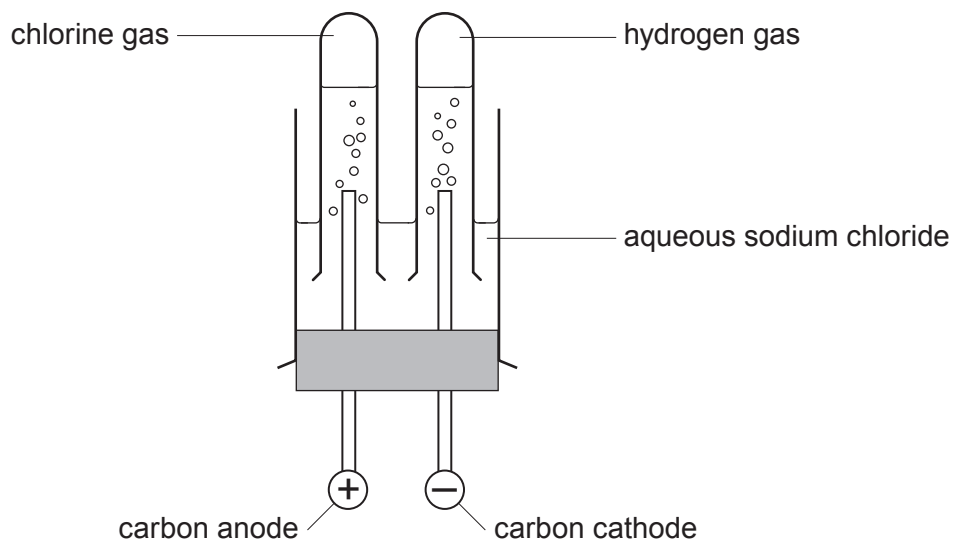
..... [1]

II. The negative ion present in the salt is CH_3COO^- .

Underline the correct formula of the salt. [1]



11. The diagram shows the laboratory apparatus that can be used to model the manufacture of aqueous sodium hydroxide (NaOH), chlorine (Cl_2) and hydrogen (H_2) from aqueous sodium chloride, NaCl .



Aqueous sodium chloride contains the ions Na^+ , Cl^- , OH^- and H^+ .



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only

[6 QER]

END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1															4 He Helium 2		
7 Li Lithium 3	9 Be Beryllium 4									11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10		
23 Na Sodium 11	24 Mg Magnesium 12									27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE – CONTINGENCY**

3410UD0-1



Z22-3410UD0-1

THURSDAY, 23 JUNE 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
 You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 4(c) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	13	
4.	11	
5.	11	
6.	7	
7.	10	
8.	8	
Total	80	

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Answer **all** questions.

1. (a) Crude oil can be separated into simpler mixtures called fractions. The table shows information about some of the main fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ –C ₄
petrol	30–75	C ₅ –C ₈
naphtha	70–180	C ₉ –C ₁₂
kerosene	180–250	C ₁₃ –C ₁₆
diesel oil	250–340	C ₁₇ –C ₂₀
lubricating oil	340–500	C ₂₀ –C ₂₄
fuel oil	490–580	C ₂₅ –C ₂₈

- (i) Decane has a boiling point of 174 °C. Give the name of the fraction which contains decane. [1]

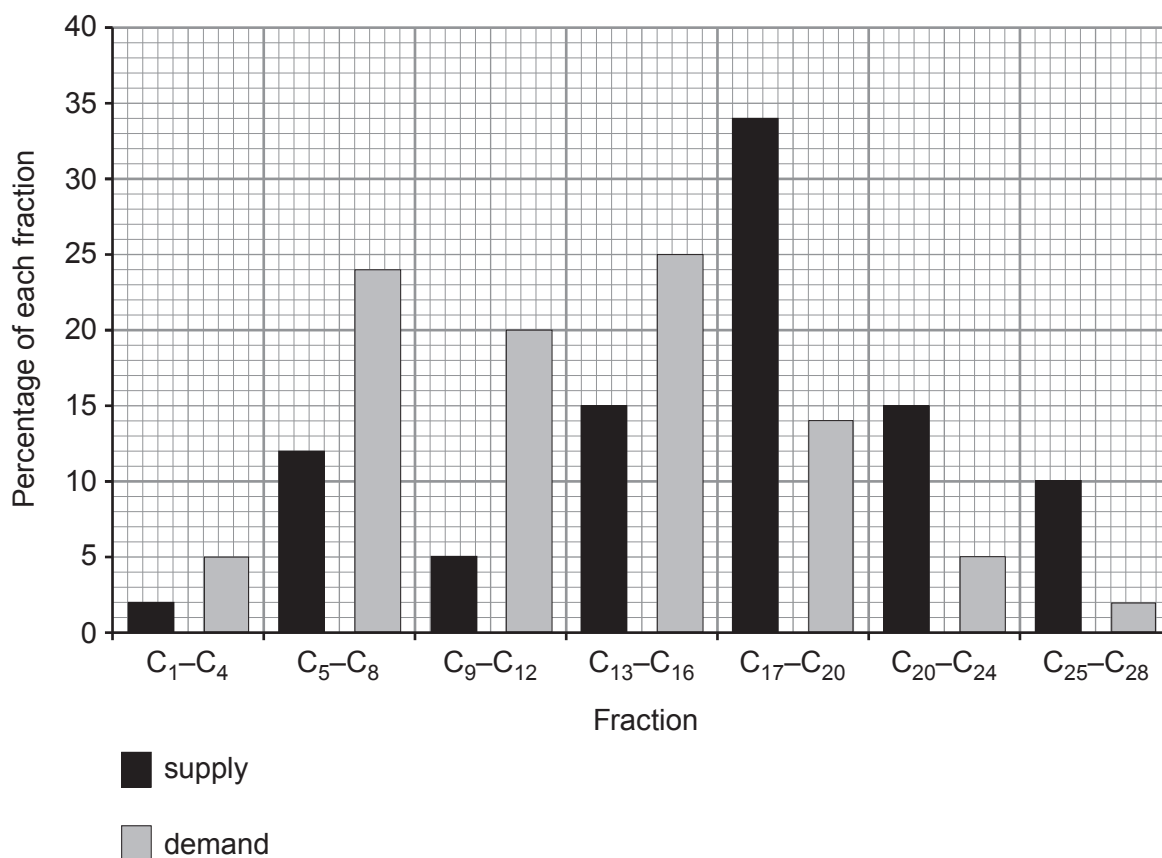
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- (ii) One alkane is found in both diesel oil and lubricating oil. Give the number of carbon atoms in this alkane. [1]

.....



- (b) The bar chart shows the supply and demand for some of the fractions obtained from crude oil.



- (i) Give the fraction where the demand is 100 % greater than the supply. [1]

.....

- (ii) Put a **tick** (✓) in the box next to the statement that best describes how the supply and demand of the fractions change as the chain length increases. [1]

supply is greater than demand for all fractions

☐

supply is greater than demand up to C₁₆ after which demand is greater than supply

☐

demand is greater than supply up to C₁₆ after which supply is greater than demand

☐

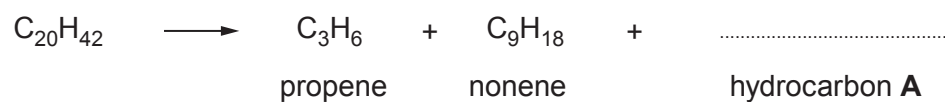
the difference between supply and demand increases up to C₁₆ after which it decreases

☐


- (iii) Oil companies have solved the problem of over-supply of some fractions using a process called cracking.

The alkane $C_{20}H_{42}$ can be cracked forming propene, nonene and **one molecule** of hydrocarbon **A**.

- I. Complete the equation for the cracking of $C_{20}H_{42}$. [1]



- II. Propene is an important raw material in the production of polypropene.

Give the name of the process used to make polypropene from propene. [1]

.....



- (c) Every year thousands of acres of forests are destroyed by wildfires. Fire-fighters use several different methods to put out this type of fire.



State **three** methods that are used to put out forest fires.

Give the part of the fire triangle being removed in each method.

Each method should refer to a different part of the fire triangle.

[3]

Method 1

Part of the fire triangle being removed

Method 2

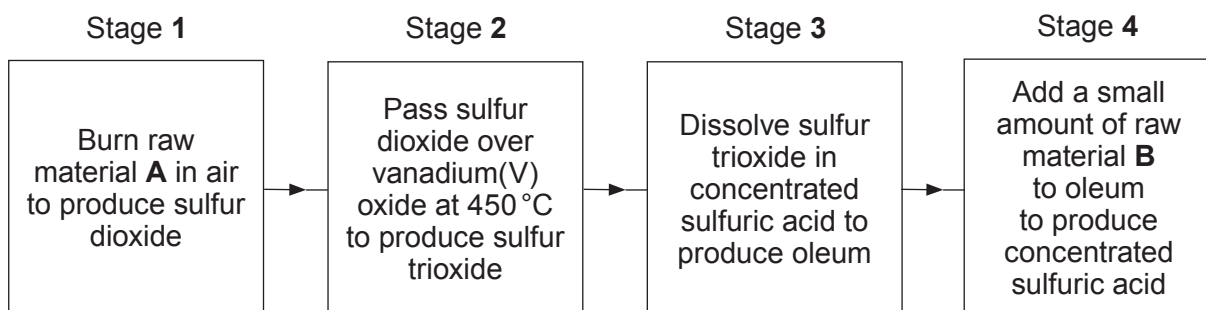
Part of the fire triangle being removed

Method 3

Part of the fire triangle being removed



2. (a) Sulfuric acid is produced by the contact process. The flow diagram shows the main stages in the process.



- (i) Give the name of raw materials **A** and **B**. [2]

A

B

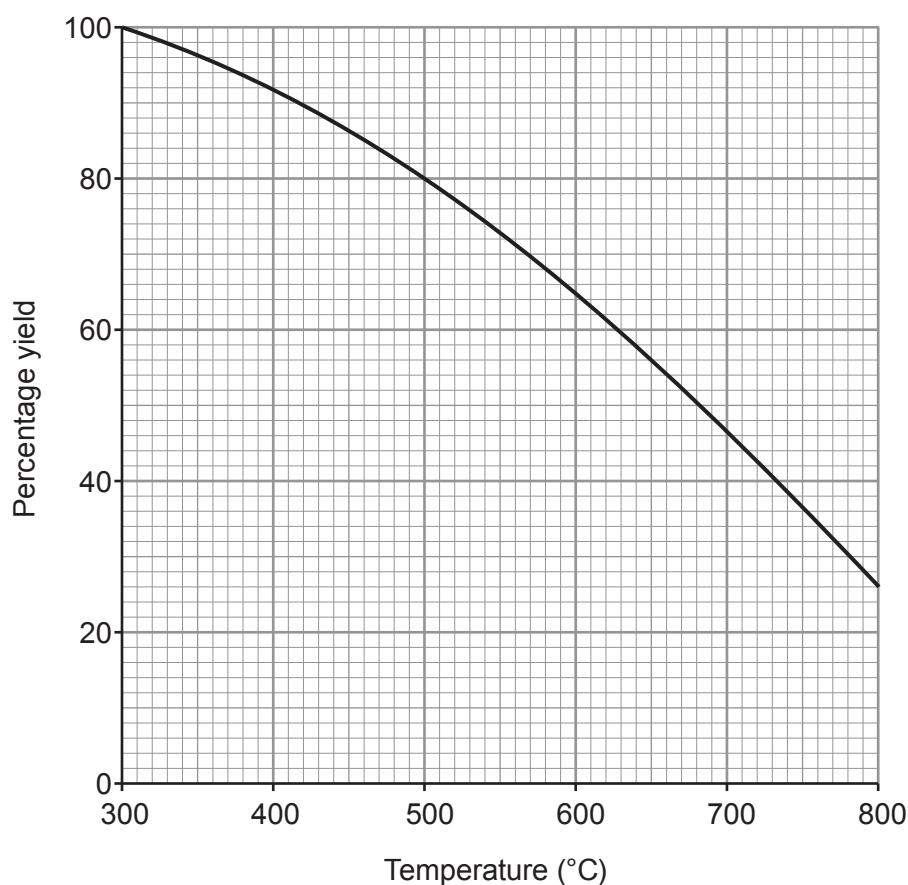
- (ii) State the purpose of vanadium(V) oxide in stage 2. [1]

.....

- (iii) Complete and balance the equation for the reaction in stage 2. [2]



- (iv) The graph shows how the percentage yield of sulfur trioxide changes with temperature between 300 °C and 800 °C.



Use the graph to answer parts I and II.

- I. State the trend in the percentage yield of sulfur trioxide as the temperature increases. [1]

.....

- II. Give the temperature **range** to be used to obtain a yield of sulfur trioxide greater than 80 %. [1]

..... to °C

- (v) One molecule of sulfur trioxide reacts with one molecule of sulfuric acid to form one molecule of oleum as the **only product**.

Complete the equation for this reaction by giving the formula of oleum. [1]



- (b) The photograph shows the exothermic reaction between concentrated sulfuric acid and sugar, $C_{12}H_{22}O_{11}$.



- (i) Name the **two** products formed. [2]

..... and

- (ii) State the type of reaction taking place. [1]

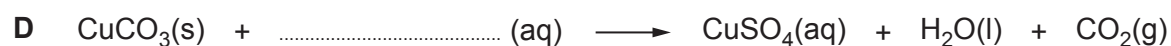
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3. (a) **A, B, C** and **D** are equations for reactions used to make salts.

(i) Complete the equation for each reaction.

[4]



(ii) Outline the **three** steps you would carry out to get a **pure** sample of silver chloride, AgCl, from the reaction mixture in equation **B**.

[2]

.....

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.....



- (b) Sodium chloride is made when sodium hydroxide reacts with hydrochloric acid.

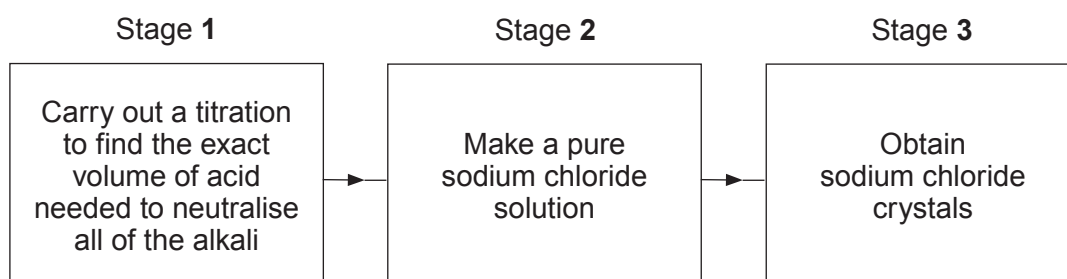


- (i) Write a balanced equation for the reaction.

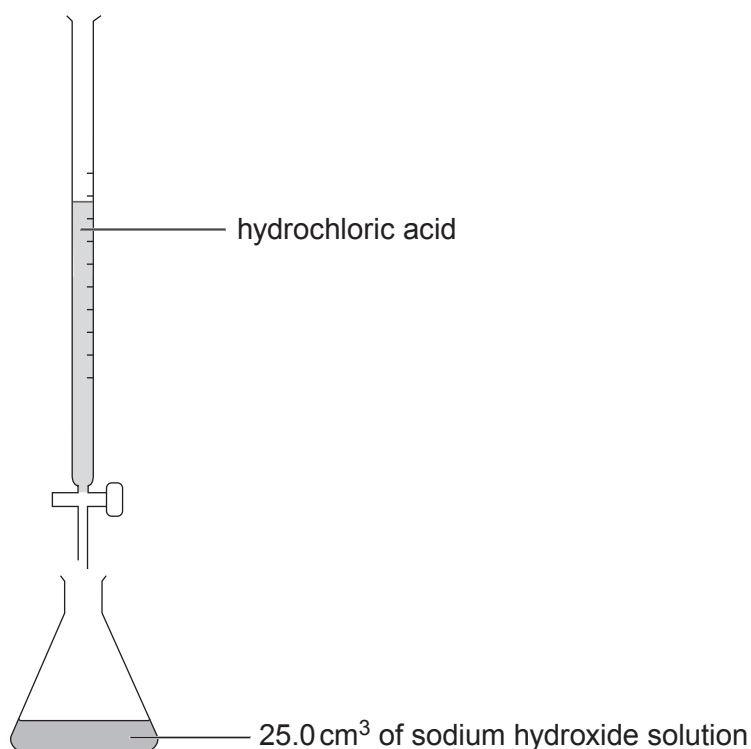
[2]

- (ii) A student was asked to prepare pure sodium chloride crystals using 25.0 cm³ of sodium hydroxide solution.

The flow diagram shows the student's plan.



- I. The student sets up the apparatus below to carry out stage 1.



Describe how the student will find the exact volume of acid needed to neutralise all of the alkali.

[2]

.....

.....

- II. The neutral solution obtained in stage 1 is not pure.

Describe what the student must do in stage 2 to make a pure sodium chloride solution.

[2]

.....

.....

.....

- III. State what the student must do in stage 3 to obtain sodium chloride crystals.

[1]

.....

.....

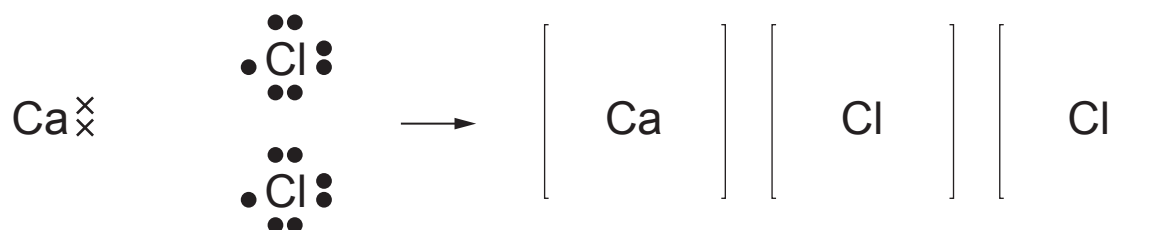


4. (a) Calcium reacts with chlorine to form the ionic compound calcium chloride.

Complete the dot and cross diagram to show the electronic changes that take place when calcium reacts with chlorine to form calcium chloride.

Show the charge on each ion.

[3]



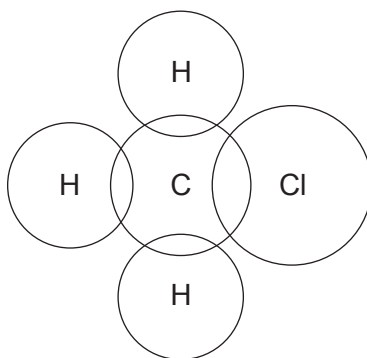
- (b) Complete the dot and cross diagram to show the bonding in a molecule of chloromethane.

[2]

hydrogen 1

carbon 2,4

chlorine 2,8,7



Examiner
only

(c) Calcium and calcium chloride have different bonding and structures.

Explain how each substance conducts electricity.

You may include diagrams in your answer.

[6 QER]

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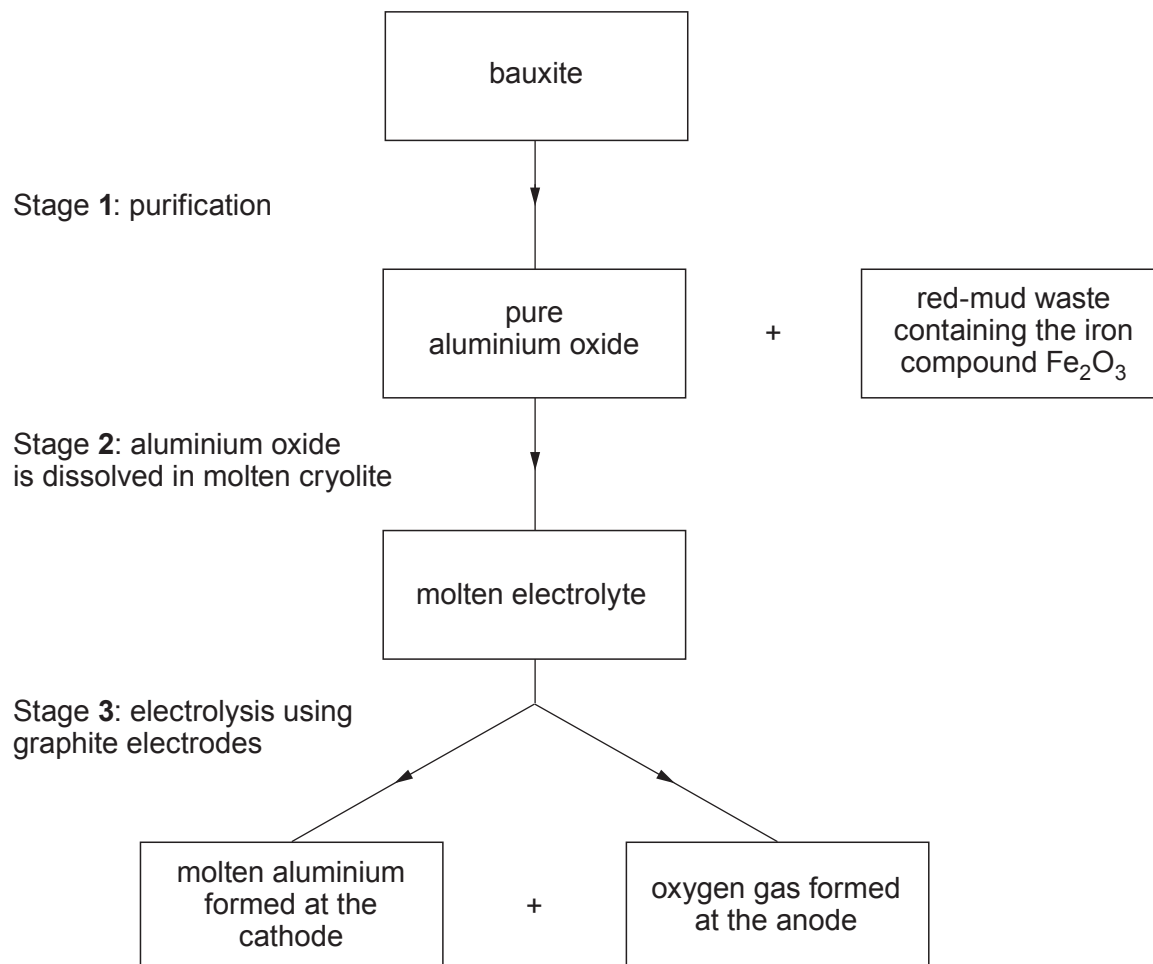
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5. (a) The flow chart shows the stages involved in the extraction of aluminium from bauxite. Bauxite is a mixture of mainly aluminium oxide and a compound of iron.



- (i) Underline the chemical name for the iron compound, Fe_2O_3 , found in the red-mud waste. [1]

iron(V) oxide

iron(II) oxide

iron(III) oxide

iron(I) oxide

- (ii) Give **one** reason why aluminium oxide is dissolved in molten cryolite in stage 2. [1]

.....
.....

- (iii) Explain why the anode is regularly replaced in stage 3. [2]

.....
.....

- (iv) Complete and balance the equation for the electrolysis of aluminium oxide. [2]



- (b) The recycling rate for aluminium drinks cans in the UK reached 72 % in 2017.

Suggest why most aluminium waste is recycled. [2]

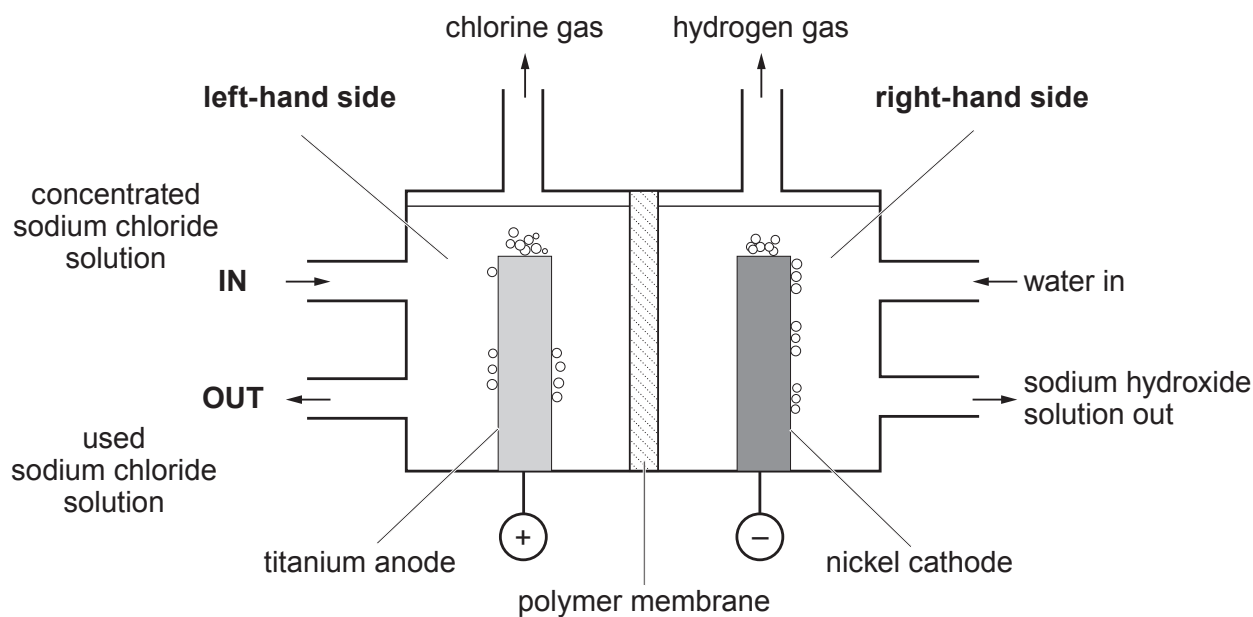
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- (c) The diagram shows a model of a membrane cell used to manufacture sodium hydroxide solution, chlorine and hydrogen by the electrolysis of aqueous sodium chloride.

The membrane cell is designed so that chlorine is kept separate from hydrogen and aqueous sodium hydroxide.

The polymer membrane only allows positive ions to pass through it. Sodium ions and hydrogen ions pass through from the left-hand side to the right-hand side of the cell.



Chlorine is an extremely corrosive substance which reacts with most metals. Titanium reacts quickly with chlorine gas but it resists attack by chlorine in aqueous conditions.



- (i) Put a **tick (✓)** in the box next to the correct list of ions found on either side of the membrane during the process. [1]

Left-hand side of the membrane	Right-hand side of the membrane
Na ⁺ Cl ⁻ H ⁺ OH ⁻	Cl ⁻ H ⁺ OH ⁻
H ⁺ OH ⁻ Cl ⁻	Na ⁺ H ⁺ OH ⁻
Na ⁺ Cl ⁻ H ⁺ OH ⁻	Na ⁺ H ⁺ OH ⁻
Na ⁺ Cl ⁻	Na ⁺ OH ⁻

☐☐☐☐

- (ii) Put a **tick (✓)** in the box next to the correct statement. [1]

titanium burns in chlorine in aqueous conditions

☐

titanium doesn't react with chlorine under any conditions

☐

aqueous conditions prevent chlorine from reacting with titanium

☐

aqueous conditions prevent chlorine from reacting with sodium chloride

☐

- (iii) Put a **tick (✓)** in the box next to the two substances formed on the right-hand side of the membrane if the sodium chloride solution were contaminated with magnesium bromide. [1]

sodium hydroxide and magnesium hydroxide

☐

sodium hydroxide and sodium bromide

☐

sodium hydroxide and magnesium bromide

☐

magnesium chloride and sodium

☐

6. (a) The table shows the formulae of the first four members of the alcohol family.

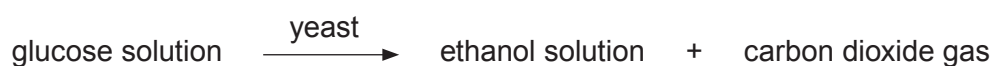
Alcohols
CH ₃ OH
C ₂ H ₅ OH
C ₃ H ₇ OH
C ₄ H ₉ OH

Give the general formula for the alcohol family.

[1]

.....

- (b) The following word equation represents the formation of ethanol by fermentation.



- (i) Complete and balance the equation for the production of ethanol and carbon dioxide from glucose.

Include the state symbols for the reactant and the products.

[3]



- (ii) Yeast contains a substance which catalyses the breakdown of glucose.

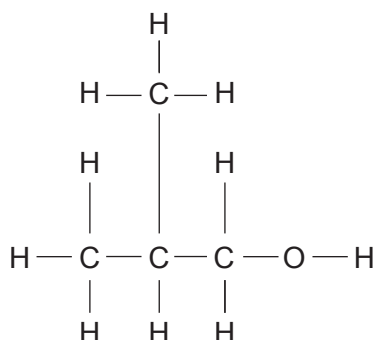
Give the **general** name of this type of catalyst.

[1]

.....



- (c) (i) C_4H_9OH has four isomers. The diagram below shows the structure of one of the isomers.



Put a **tick (✓)** in the box next to the name of this isomer.

[1]

butan-1-ol

☐

2-methylpropan-1-ol

☐

butan-2-ol

☐

2-methylpropan-2-ol

☐

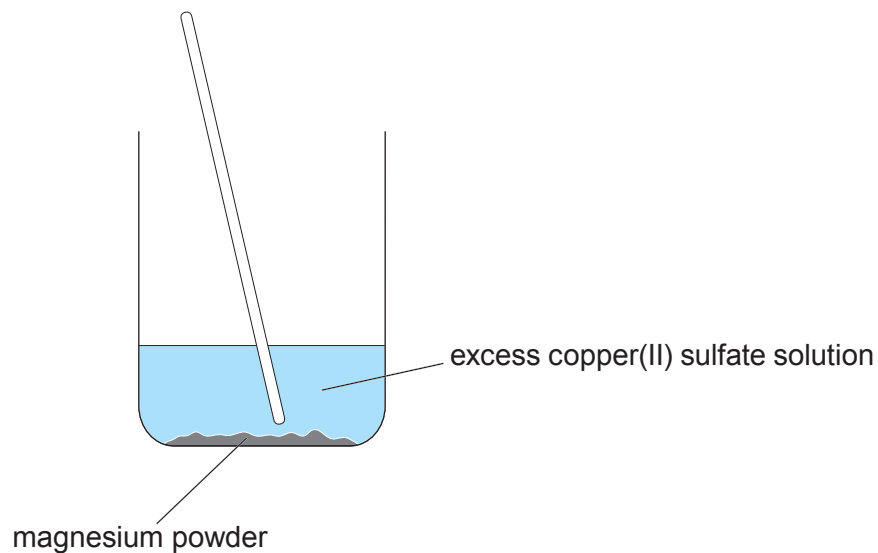
- (ii) Pentan-3-ol is one of the isomers of $C_5H_{11}OH$.

Draw the structure of pentan-3-ol.

[1]



7. (a) A student investigated the mass of copper formed when magnesium powder was added to excess copper(II) sulfate solution.



The student added four different masses of magnesium powder to separate 50 cm³ samples of copper(II) sulfate solution.

The table shows the results obtained.

Mass of magnesium added (g)	Mass of copper formed (g)
0.00	0.00
0.05	0.13
0.10	0.26
0.15	0.39
0.20	0.52

- (i) Give **one** observation that would show that copper(II) sulfate is always in excess in this investigation. [1]

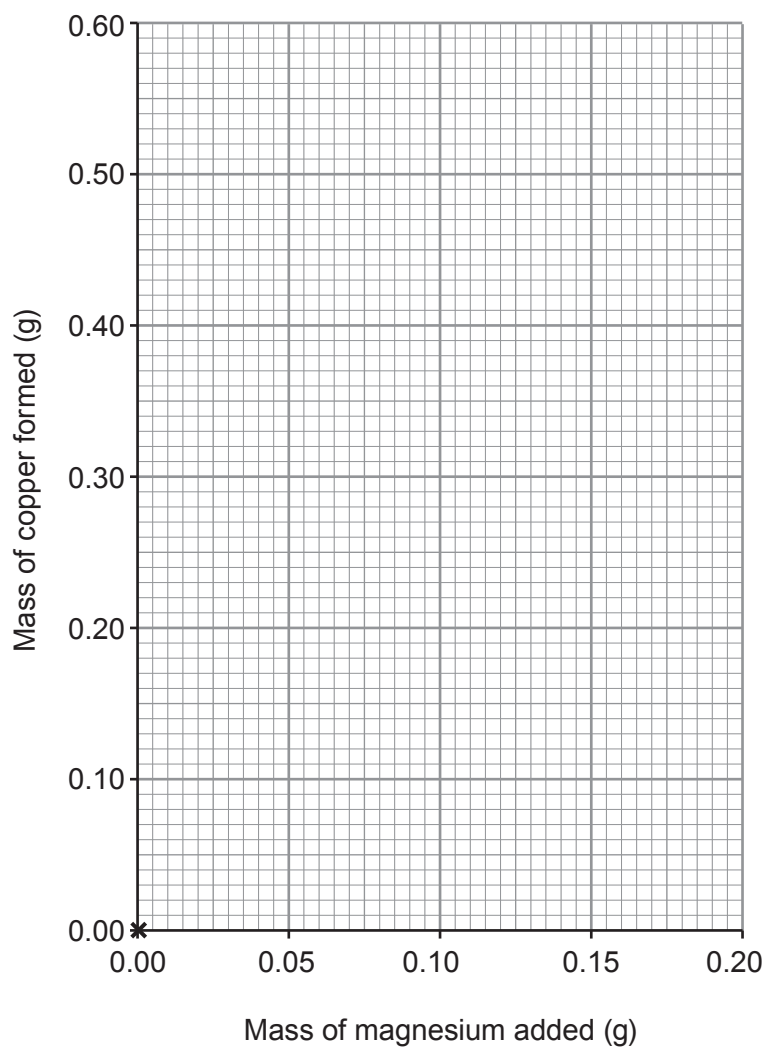
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- (ii) Plot the results from the table on the grid and draw a suitable line. [3]

The point (0,0) has been plotted for you.



- (iii) I. Use the graph to find the mass of magnesium needed to form 0.18 g of copper. [1]

..... g

- II. The student stated that 0.78 g of copper would be formed when 0.30 g of magnesium is used. Suggest how she arrived at this value. [1]

.....

.....

.....



- (b) To ensure that the copper(II) sulfate solution was always in excess the technician dissolved 3.19 g of CuSO_4 in 500 cm^3 of water.

- (i) Calculate the number of moles of copper(II) sulfate in 3.19 g. [2]

$$A_r(\text{O}) = 16 \qquad A_r(\text{S}) = 32 \qquad A_r(\text{Cu}) = 63.5$$

Number of moles = mol

- (ii) Use the equation below to calculate the concentration of the copper(II) sulfate solution in mol/dm^3 . [2]

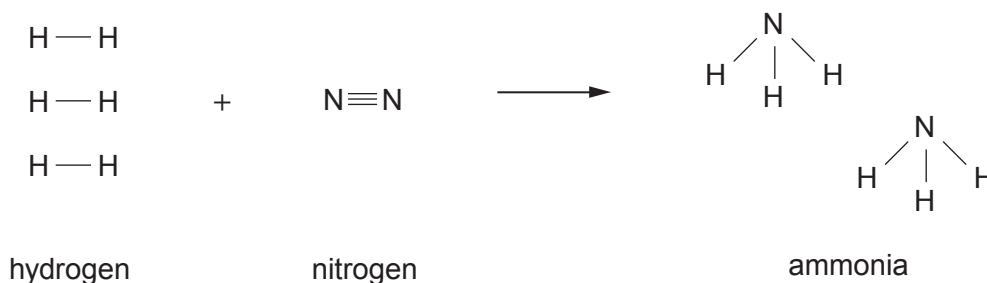
$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{number of moles}}{\text{volume (dm}^3\text{)}}$$

Concentration = mol/dm^3



8. Ammonia is manufactured from hydrogen and nitrogen in the Haber process.

- (a) The equation shows the bonds which are broken and the bonds which are formed in the production of ammonia.

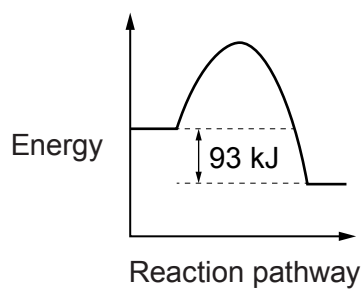


The bond energy of a H—H bond is 436 kJ. The energy needed to break **all** the bonds in the **reactants** is 2253 kJ.

- (i) Calculate the energy needed to break a N≡N bond. [2]

Energy = kJ

- (ii) The diagram below is the energy profile for the reaction.



Calculate the energy released when **all** the bonds in the ammonia molecules are formed. [1]

Energy = kJ



- (b) The table below shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)			
	200	300	400	500
	Percentage yield of ammonia			
100	81.7	52.5	25.2	10.6
200	89.0	66.7		18.3
400	94.6	79.7	55.4	31.9
1000	98.3	92.6	79.8	57.5

- (i) Describe the relationship between the percentage yield of ammonia and the temperature.

[1]

- (ii) **Circle** the most likely percentage yield of ammonia if the process were carried out at 200 atm and 400 °C.

[1]

20 %

30 %

40 %

50 %



- (c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

Compound	Flame test	Add NaOH(aq)	Add HCl(aq)	Add BaCl ₂ (aq)	Add AgNO ₃ (aq)
A	no colour	pungent gas given off turns damp red litmus paper blue	no reaction	no reaction	white precipitate forms
B	green flame	blue precipitate forms	no reaction	white precipitate forms	no reaction
C	lilac flame	no reaction	gas given off turns limewater milky	no reaction	no reaction

Use the information provided to identify compounds **A**, **B** and **C**.

[3]

A

B

C

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

0

[illegible]

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410UB0-1



S23-3410UB0-1

MONDAY, 22 MAY 2023 – MORNING**CHEMISTRY – Unit 2:****Chemical Bonding, Application of Chemical Reactions and Organic Chemistry****HIGHER TIER**

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 6(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	9	
4.	7	
5.	11	
6.	8	
7.	9	
8.	10	
9.	6	
Total	80	

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Answer **all** questions.

1. (a) A student made some copper(II) sulfate crystals by reacting copper(II) carbonate powder with sulfuric acid using the following method.

Stage 1 Measure 50 cm³ of sulfuric acid into a beaker.

Stage 2 Add copper(II) carbonate powder, one spatula at a time, until all the acid has reacted.

Stage 3 Filter the mixture.

Stage 4 Obtain crystals from the solution.

- (i) State how you would carry out Stage 4 to get the largest possible crystals. [1]

.....

.....

- (ii) Crystals of copper(II) sulfate could also be made using copper(II) oxide powder instead of copper(II) carbonate powder. State and explain how the observations in Stage 2 would be different. [2]

.....

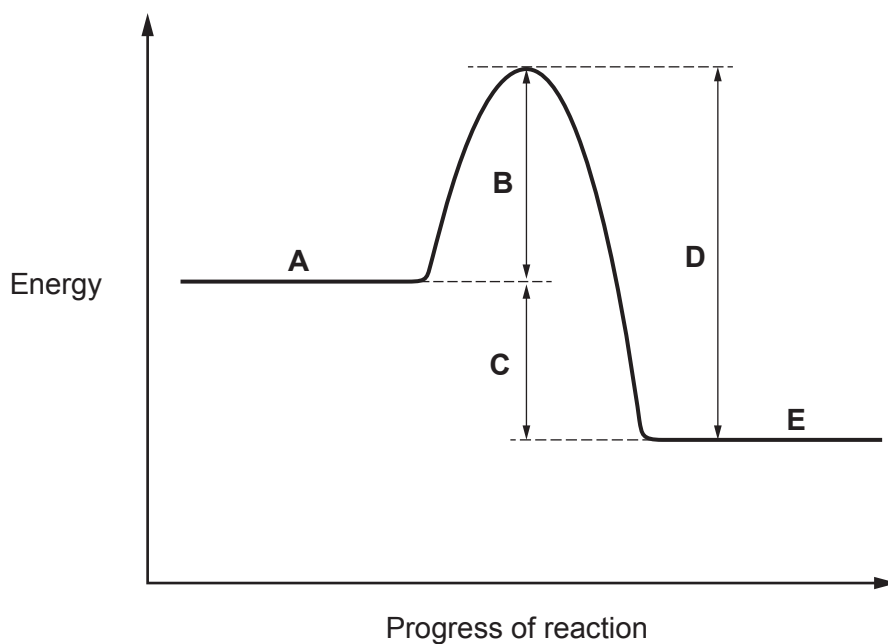
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- (iii) Complete the symbol equation for the reaction between copper(II) oxide and sulfuric acid. Copper(II) sulfate is one of the products. [2]



(b) The diagram shows an energy profile for a reaction.



(i) Give the **letter** that represents each of the following parts of the energy profile. [2]

Part of the energy profile	Letter
energy change for the reaction	
energy of the reactants	
activation energy of the reaction	

(ii) Give the meaning of the term activation energy. [1]

.....

.....

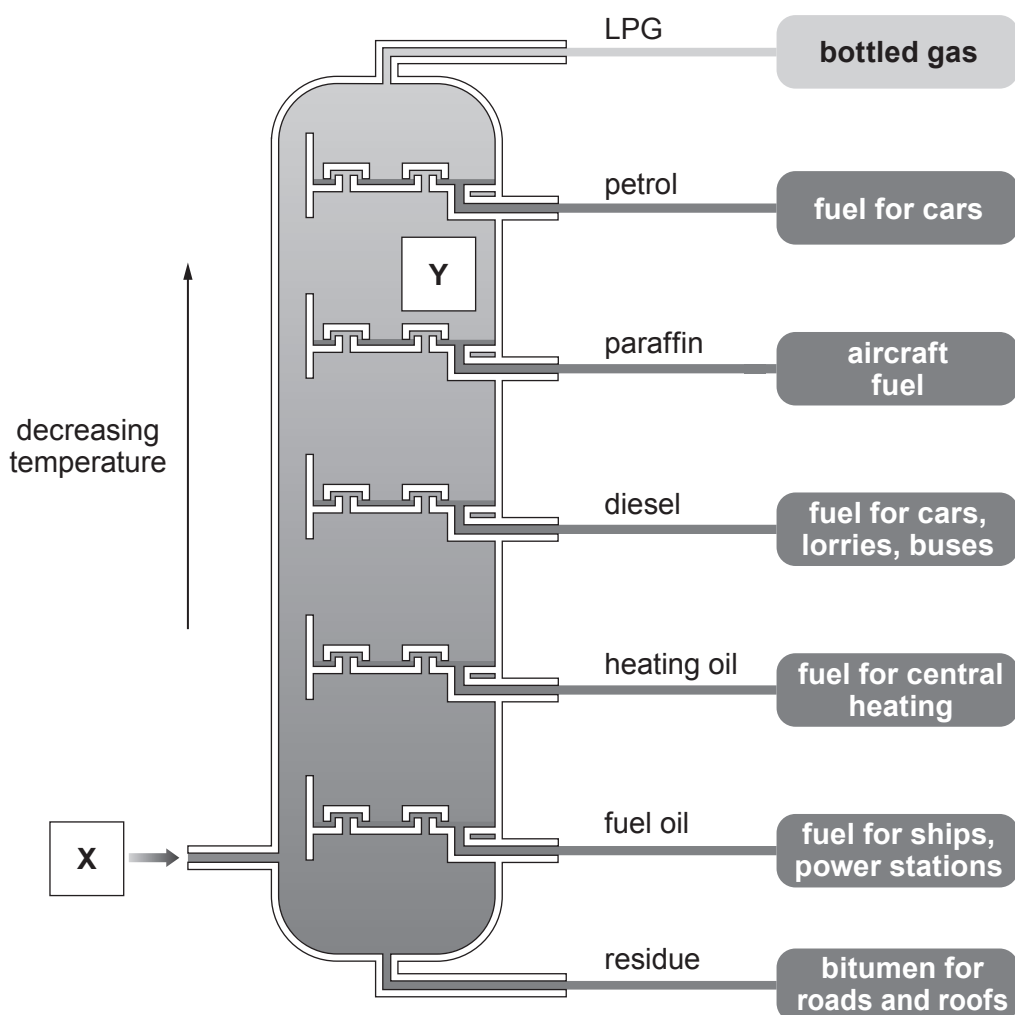
(iii) State how the energy profile shows that this is an exothermic reaction. [1]

.....

.....



2. (a) Crude oil is separated into mixtures of hydrocarbon compounds in the process of fractional distillation. Many of these fractions are used as fuels.



- (i) Name the changes of state happening at **X** and at **Y**.

[1]

X

Y

- (ii) Explain why different fractions are formed at different levels.

[2]

.....

.....

.....

.....



- (iii) A hydrocarbon fuel was burned and used to heat 100g of water. The water temperature rose from 18.5 °C to 38.2 °C.

Use the equation below to calculate the amount of energy released by this fuel. Give your answer to **two** significant figures. [3]

$$\text{energy (J)} = \text{mass of water (g)} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}$$

Energy = J

- (b) The products of fractional distillation can undergo a process called cracking to produce smaller, more useful hydrocarbons.

- (i) Complete the equation for the cracking of $\text{C}_{16}\text{H}_{34}$. [1]



- (ii) State the **two** conditions used for cracking. [1]

.....

- (iii) The molecule with the formula C_2H_4 is an unsaturated hydrocarbon.

Give the meaning of the term unsaturated. [1]

.....

.....

- (iv) State why there is a high demand for each of the following products of the cracking reaction. [2]

octane / C_8H_{18}

ethene / C_2H_4



3. (a) The table shows information about some organic compounds.

Name	Molecular formula	Structure	Homologous series
ethanol	C_2H_5OH		alcohols
ethanoic acid	CH_3COOH	<pre> H H — C — C // \ H O O — H </pre>	
.....	C_3H_6	<pre> H H H H — C = C — C — H H H H </pre>	alkenes

(i) **Complete the table.**

[3]

(ii) The molecular mass of an alkene is 98. Give the molecular formula of this alkene.

[1]

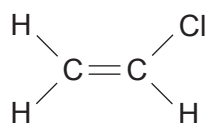
$$A_r(C) = 12$$

$$A_r(H) = 1$$

Molecular formula

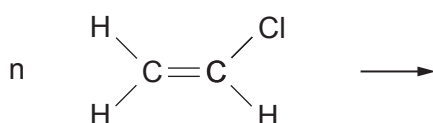


- (b) Polyvinyl chloride (PVC) is formed from the following monomer in a polymerisation reaction.



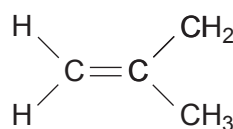
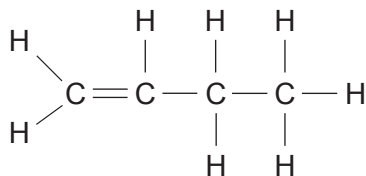
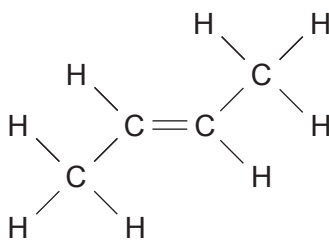
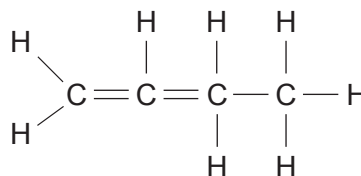
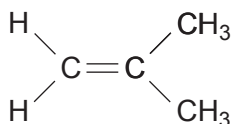
Complete the equation for the polymerisation reaction.

[2]



- (c) A student was asked to draw structures for isomers with the molecular formula C_4H_8 . She drew the following diagrams. Circle **all** the **correct** structures.

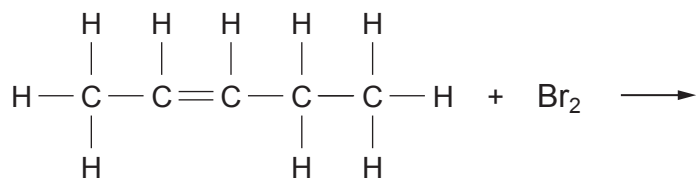
[2]



(d) Pent-2-ene reacts with bromine water in a similar manner to ethene.

Complete the equation for the reaction of pent-2-ene with bromine by drawing the structure of the product.

[1]



Examiner
only

9



4. (a) The table shows the results of an experiment in which zinc and lead powders were added separately to solutions of sodium chloride and iron(II) chloride.

	Sodium chloride solution	Iron(II) chloride solution
zinc	no change	colour change
lead	no change	no change

Use the results to place the four metals in order of reactivity. [1]

Most reactive
.....
.....
Least reactive
.....



- (b) A student investigated metal reactivity using a different method.

He added 10.0 g of magnesium powder in 2.0 g portions to 50 cm³ of zinc chloride solution and recorded the temperature of the mixture after each addition. He repeated the experiment with aluminium powder and again with copper powder.

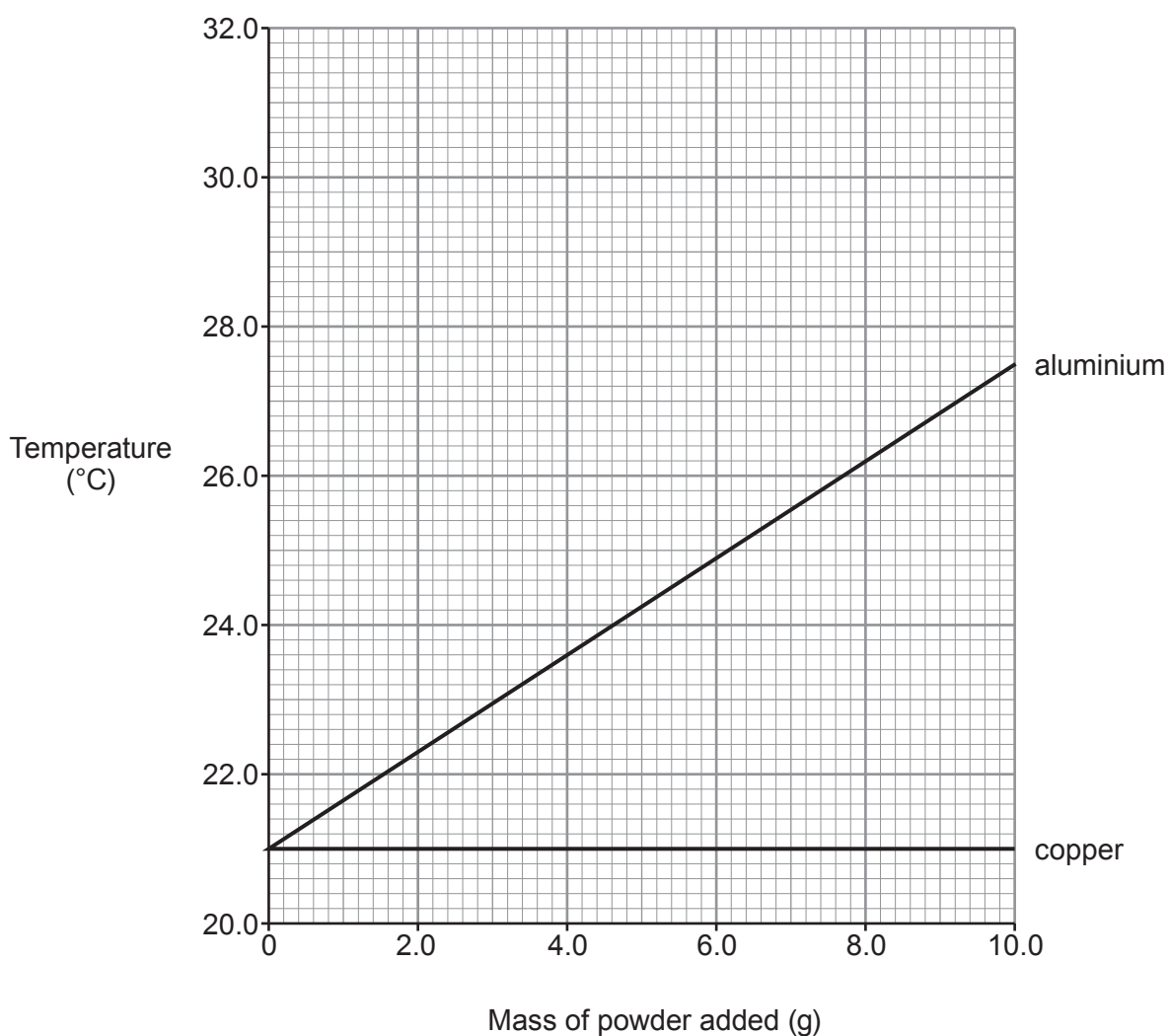
The results for magnesium powder are shown below.

Mass of magnesium powder added (g)	Temperature (°C)
0	21.0
2.0	23.2
4.0	25.1
6.0	27.5
8.0	29.7
10.0	31.6

The results for aluminium and copper have been plotted on the grid opposite.



- (i) Plot the results for magnesium on the grid. Draw a suitable line. [3]



- (ii) State the conclusions that can be drawn about the reactivities of magnesium, aluminium, copper and zinc. Give your reasoning. [3]

.....

.....

.....

.....



5. (a) (i) Draw a dot and cross diagram to show the bonding in a carbon dioxide molecule. [2]

carbon (2,4)

oxygen (2,6)

- (ii) Explain why carbon dioxide has a low boiling point. [2]

.....

.....

.....

.....



- (b) (i) Draw a diagram to show the electronic changes that take place during the formation of magnesium oxide. Include the charges on the ions formed. [2]

magnesium (2,8,2)

oxygen (2,6)

- (ii) Explain why magnesium oxide has a higher melting point than sodium chloride. [2]

.....

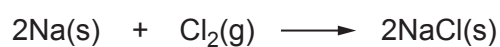
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- (iii) Sodium chloride is produced by the reaction of sodium with chlorine.



In a reaction using 0.080 mol of sodium, 4.12 g of sodium chloride was produced.

Calculate the percentage yield of this reaction. [3]

$$A_r(\text{Na}) = 23$$

$$A_r(\text{Cl}) = 35.5$$

Percentage yield = %



- (b) State how adding sodium hydroxide solution allows solutions containing iron(II) and iron(III) ions to be identified. Give the observations made for both solutions. [2]

.....

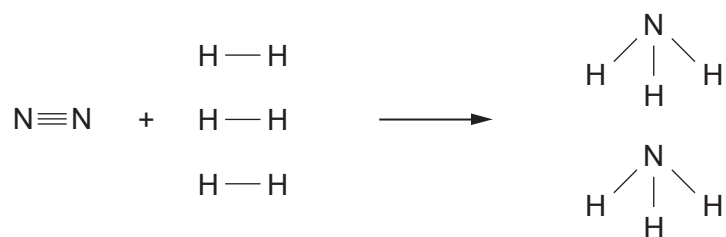
.....

Examiner
only

8



7. (a) The formation of ammonia can be represented by the following equation.



- The total amount of energy released in making the bonds in the products is 2340 kJ
- The total amount of energy released in making the bonds in the products is 94 kJ **more than** the total energy used in breaking the bonds in the reactants
- The amount of energy used to break the $\text{N} \equiv \text{N}$ bond is 941 kJ

Use this information to calculate the energy used to break **one** $\text{H} - \text{H}$ bond.

[3]

Energy = kJ



- (b) The Haber process used to make ammonia is usually carried out at a temperature of around 400 °C. Explain why this is the optimum temperature.

[2]

A higher temperature is not used because

.....

A lower temperature is not used because

.....

- (c) Ammonia is used in the production of fertilisers such as ammonium nitrate.

Give the balanced symbol equation for the reaction between nitric acid and ammonia to form ammonium nitrate.

[2]

.....

- (d) Eutrophication is a problem caused by fertilisers being washed into waterways leading to overgrowth of plants. Describe how eutrophication leads to the death of aquatic organisms.

[2]

.....

.....

.....

.....



8. (a) According to the Journal of the British Dental Association, the increase in consumption of fruit juices and other acidic drinks is believed to be one of the leading causes of dental erosion in children and adolescents.

Many fruit juice drinks contain more than one type of acid. Citric acid can be used as a natural preservative and provides a sour taste. Ascorbic acid (vitamin C) is a water-soluble vitamin that must be consumed regularly to ensure proper body function. Citrus fruits, as well as tomatoes and other fresh vegetables, are good sources of vitamin C.

The table below shows information about the content of different fruit juice drinks.

Fruit juice drink	pH	Water (%)	Citric acid (%)	Ascorbic acid (mg/100g)	Sugar (%)
lime	2.2	76.4	4.80	29	1.68
lemon	2.4	81.3	4.60	53	1.80
grapefruit	3.0	90.3	1.35	38	2.34
orange	3.6	87.4	0.96	50	4.85
tangerine	3.8	85.7	0.74	31	7.89

- (i) Tick (✓) the **two** conclusions that can be drawn from the information.

[2]

as pH increases, citric acid content decreases and sugar content increases	
as acidity decreases, ascorbic acid content decreases and water content decreases	
tomatoes are a good source of vitamin C and citric acid	
citrus fruits contain ascorbic acid and a natural preservative	

- (ii) **Use information from the table** to suggest why citric acid content has a much greater effect on the pH than ascorbic acid content.

[1]

.....

.....



- (b) The concentration of a solution of sodium hydroxide can be determined by titrating with sulfuric acid. The equation for the reaction is shown.



- (i) Give the **ionic** equation for the formation of water in any neutralisation reaction. Include state symbols. [2]

-
- (ii) 21.0 cm³ of sulfuric acid with a concentration of 0.350 mol/dm³ neutralised 25.0 cm³ of the sodium hydroxide solution.

- I. Calculate the number of moles of sulfuric acid used in the reaction. [1]

Number of moles = mol

- II. Calculate the concentration of the sodium hydroxide solution. [2]

Concentration = mol/dm³



- (iii) During a similar titration reaction, 0.36 g of water was produced. Calculate the number of **molecules** of water produced in this reaction.

Give your answer in **standard form**.

[2]

$$A_r(\text{H}) = 1$$

$$A_r(\text{O}) = 16$$

$$\text{Avogadro's constant} = 6.0 \times 10^{23}$$

Number of molecules =

10



9. Aluminium metal is produced industrially using electrolysis, due to the relatively high stability of aluminium oxide.

The overall equation for the reaction is



- (a) Describe, in terms of electrons, the reduction and oxidation occurring during the electrolysis.

[2]

.....

.....

.....

.....



- (b) (i) Aluminium ore contains 36 % aluminium oxide, Al_2O_3 .

Calculate the mass of aluminium oxide in 500 tonnes of the ore.

[1]

Mass = tonnes

- (ii) Calculate the mass of aluminium metal that could be produced from this mass of aluminium oxide.

[3]

$$A_r(\text{Al}) = 27$$

$$A_r(\text{O}) = 16$$

Mass = tonnes

6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1 4 He Helium 2																	
7 Li Lithium 3	9 Be Beryllium 4																
23 Na Sodium 11	24 Mg Magnesium 12																
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

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